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Dissertation

COVARIATION & SALIENCE IN LINGUISTIC CONTACT:

A SOCIOPHONETIC STUDY OF LIQUID VARIATION IN BOSTON SPANISH

by

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Nature does not hurry, yet
everything is accomplished.

Lao Tzu

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ABSTRACT

This dissertation examines how Spanish-speaking Bostonians of Puerto Rican and Dominican origin navigate linguistic variation and lectal coherence, focusing on the role of salience in shaping the outcomes of linguistic contact. Speakers do not adopt, resist, or modify linguistic features at random—social and cognitive factors shape how variation unfolds in contact settings. While prior research has shown that some features vary in their susceptibility to change in bilingual environments, the extent to which salience influences this process remains underexplored. By analyzing both low-salience features (e.g., *subject pronoun expression*, *subject placement,* and *filled pauses*) and high-salience features (e.g., *coda /s/* and *liquid variation*), this study explores how speakers engage with variation in response to social and linguistic pressures in a dynamic contact setting.

Using sociolinguistic interviews from 22 Spanish speakers in Boston, the study employs quantitative variationist methods to analyze the distribution and interaction of multiple variables. The results show that low-salience features exhibit modest, yet, measurable and systematic structural convergence with English grammatical norms, reflecting bilingual optimization strategies. In contrast, high-salience features provide speakers with greater social and stylistic flexibility, allowing them to deploy these features in ways that align

with their identity, ideology, and interactional goals. In some cases, such as with rhotics, speakers maintain a broader sense of Caribbean linguistic identity—even after extended residence in the mainland U.S.—using these features as shibboleths of national identity. For example, the lateralization of the tap continues to differentiate Puerto Rican and Dominican speech styles, serving as a salient marker of regional affiliation. However, other high-salience features, such as laterals, appear to carry less social weight, making them more susceptible to contact-induced restructuring or stylistic reinterpretation.

To examine how these patterns of variation intersect, the study incorporates a covariation analysis that investigates whether and how the five variables pattern together—particularly in relation to speakers' exposure to English and broader contact-induced pressures. This analysis reveals a fundamental asymmetry: lower-salience features tend to shift in tandem, reflecting a broader structural realignment shaped by bilingual optimization. Higher-salience features, by contrast, do not show the same kind of coordinated movement. Instead, they remain sites of sociolinguistic agency, affording speakers the flexibility to reinforce, neutralize, or reinterpret these forms in ways that serve social and identity-related goals. These findings contribute to sociolinguistic theory by demonstrating that salience is not merely an inherent property of linguistic variables but a dynamic, ideologically mediated phenomenon—one that shapes how and why speakers retain, modify, or reconfigure particular linguistic features in language- and dialect-contact settings.

Moreover, the current research advances understanding of linguistic agency, identity, and change in multilingual communities by conceptualizing salience as a continuum rather than a dichotomy. By framing linguistic features along a salience continuum, the study reveals how different variables interact with social forces in contact settings, offering insights into the ways speakers navigate variation to manage group affiliations and respond to sociolinguistic pressures. Additionally, by incorporating liquid variation into the study of salience and covariation, this dissertation broadens the empirical and theoretical scope of the study of Spanish in the United States, shedding new light on the ways bilingual speakers actively engage with variation.

Contents

~~List of Tables xv~~

~~List of Figures xviii~~

1 Introduction 1

1.1 Research Questions and Objectives 2

1.2 Context and Scope 3

1.3 Hypotheses and Contributions 5

1.4 Structure of the Dissertation 6

1.5 Significance of the Study 7

~~2 Previous Literature & Motivation for the Current Study 9~~

~~2.1 Hispanic Linguistics in Contact: A Comparative Variationist Framework . . 9~~

~~2.2 Salience in Sociolinguistics . . . . 13~~

~~2.3 Covariation and Coherence . . . . 19~~

~~2.4 The Present Study . . . . 21~~

~~2.4.1 Liquid Variation . . . . 27~~

~~2.4.2 Additional Variables: An Introduction . . . . 31~~

~~2.5 Salience of the Variables in the Study . . . . 39~~

~~3 Data & Methods 44~~

~~3.1 Data Collection . . . . 45~~

~~3.1.1 Data Preparation and Coding . . . . 46~~

~~3.2 Study Design . . . . . 46~~

~~3.2.1 Dependent Variable: Liquid Variation . . . . . 47~~

~~3.2.2 Segmentation Procedure for Liquid Tokens . . . . . 50~~

~~3.2.3 Linguistic Factors . . . . . 54~~

~~3.2.4 Social Factors and Language Contact Measures . . . . . 57~~

~~3.3 Statistical Model Selection Process . . . . . 59~~

~~**4 Speaker Profiles: Language Use and Attitudes** **62**~~

~~4.1 Speaker Selection Criteria and Description . . . . . 62~~

~~4.2 Language Use & Attitudes Toward Dialectal Variation . . . . . 64~~

~~4.3 Saying and Not Saying: The Role of Salience in Speaker Commentary . . . 69~~

~~4.3.1 Explicit Descriptions of Variation . . . . . 70~~

~~4.4 Evaluating Dialects: Attitudes and Social Meaning . . . . . 80~~

~~4.5 Dialect Perceptions and Identity: Key Takeaways . . . . . 85~~

5 Results: An Examination of Liquids **87**

5.1 General Overview of the Variation in the Dependent Variable 88

5.2 Analysis of Tap, Trill, and Lateral 91

5.2.1 Statistical Analysis of Variation in the Tap — /ɾ/ 92

5.2.2 Tap Subset: Laterals Mismatch 98

5.2.3 Statistical Analysis of Variation in the Trill — /r/ 102

5.2.4 Trill Subset: [r̥] Mismatch 105

5.2.5 Statistical Analysis of Variation in the Lateral — /l/ 108

5.2.6 Summary of Findings Across Liquid Models 115

6 Covariation and Coherence: Relationships between Linguistic Variables 119
6.1 Overview of Empirical Patterns 121
6.2 Variation in Lower Salience Variables in a Contact-Setting 124
6.2.1 Zooming In: Low-Salience Features Among 22 Speakers 125
6.3 Salient Features as Tools for Speaker Identity 128
6.4 Intermediate Conclusion: Salience and Covariation 133
7 Discussion & Conclusions: Linguistic Salience and Contact-Induced Change 135
7.1 Linguistic and Social Conditioning of Liquid Variation 136
7.1.1 Taps: Strong Internal Conditioning, Dialect-Based Social Patterning 137
7.1.2 Trills: Bilingual Contact as a Primary Driver 138
7.1.3 Laterals: Contact-Induced Restructuring and Dialect Mixing 139
7.2 Covariation in Contact Settings: Structural Convergence vs. Sociolinguistic Indexing 141
7.2.1 The Social Power of Salient Variables: In-Group Membership and Indexicality 143
7.2.2 Contextualizing the Findings: A Non-Generalizable Pattern 148
7.3 Conclusion and Future Directions 150
7.3.1 Final Thoughts 152
~~A Speaker Demographics 154~~
~~A.1 Summary Statistics of Key Demographic Variables . . . . 155~~
~~A.2 Rates in Linguistic Features Across Speakers . . . . 156~~
~~A.3 Participant Demographics . . . . 156~~
~~A.4 Speaker Demographic Profiles . . . . 157~~

~~A.4.1 Dominican Republic . . . . 158~~
~~A.4.2 Puerto Rico . . . . 168~~
~~**B Coding Manual 178**~~
~~**C Spanish in Boston Questionnaire (Spanish) 193**~~
~~**D Spanish in Boston Questionnaire (English) 198**~~
~~**Bibliography 203**~~
~~**Curriculum Vitae 220**~~

1 Introduction

On a typical day, a Spanish-speaking Bostonian of Puerto Rican or Dominican origin might seamlessly navigate multiple linguistic environments. At home, they speak in rapid, Caribbean-inflected Spanish, dropping final /s/ and occasionally lateralizing a tap, just like their parents and grandparents do. At work or school, their Spanish shifts—perhaps they hyperarticulate their rhotics or pronounce syllable-final /s/ more frequently, subtly adjusting to perceptions of prestige or clarity. When chatting with bilingual friends, they code-switch fluidly, peppering their Spanish with English borrowings and sometimes even altering the structure of their sentences to align more closely with English syntax.

Yet, not all linguistic features are treated the same. Some, like pronoun presence or subject placement, tend to escape speakers' conscious awareness and rarely become the subject of overt commentary or concern. Others—like the realization of rhotics or coda /s/—are highly noticeable and frequently referenced as markers of group identity or dialectal difference. A Puerto Rican speaker might velarize their trill to distinguish themselves from a Dominican peer, while a Dominican speaker might lean into their own dialectal patterns as an assertion of identity. These choices are not random. Rather, they reflect asymmetries in linguistic awareness and the social meanings attached to particular forms.

This dissertation investigates these dynamics, focusing on how Spanish-speaking Bostonians navigate linguistic variation in a bilingual setting. Specifically, it examines how differential salience—the degree to which linguistic features are noticed, socially meaningful, and available for speaker manipulation—shapes patterns of variation and covariation. While some variables may serve as markers of identity and be deployed strategically, others may shift more subtly in response to broader structural pressures, such as those that arise in contexts of bilingualism. Recent work in sociolinguistics has shown that linguistic variables often do not vary in isolation: instead, they exhibit patterned relationships with one another—a phenomenon known as covariation. These patterns may reflect shared social meanings, stylistic coherence, or cognitive pressures associated with bilingual speech.

By analyzing variables at the lower end of the salience spectrum, which tend to shift in response to bilingual optimization, alongside variables at the higher end of the salience spectrum, which offer speakers greater opportunities for social signaling and stylistic agency, this study sheds light on the mechanisms through which linguistic features co-vary, converge, or resist change in contact settings.

1.1 Research Questions and Objectives

This study builds on prior research on Spanish in the United States by incorporating variation in the production of liquids into the discussion of the themes of salience and covariation, expanding the range of variables examined at the higher end of the salience spectrum.

The dissertation sets out to answer the following research questions:

1. How do linguistic and social factors shape variation in the production of liquids in Spanish?
2. Do the data support the hypothesis that features at the lower end of the salience continuum, such as *subject personal pronouns*, *subject placement*, and *filled pauses*, pattern

differently than those at the higher end, such as *coda /s/* and *liquid variation*?

(a) In this contact situation, do speakers' treatment of the features at the lower end of the salience continuum follow a pattern of change?

(b) Do speakers treat coda /s/ and liquid variation differently than features at the lower end of the salience continuum?

These questions inform the methodological and theoretical approach of the dissertation, linking individual linguistic variables to broader patterns of potential language change, identity construction, and contact-induced restructuring.

1.2 Context and Scope

Boston's Puerto Rican and Dominican communities provide an ideal setting for exploring these questions. As bilingual speakers immersed in an English-dominant environment, they are constantly navigating multiple linguistic boundaries—not only between Spanish and English, but also between distinct varieties of Spanish. These speakers make choices about which features to retain, adapt, or downplay, often in response to shifting social contexts. For some, linguistic behavior reflects an orientation toward maintaining dialectal distinctiveness; for others, accommodation or leveling may be preferred, whether consciously or not. These choices are shaped by a range of factors, including migration history, community networks, and ideological perceptions of language and identity.

Data for this study come from the *Spanish in Boston Project*, a large-scale variationist survey designed to provide insight into the linguistic practices and attitudes of Spanish-speaking Bostonians, with an emphasis on the outcomes of language and dialect contact. This broader project has investigated a set of four sociolinguistic variables: (i) subject pronoun expression, (ii) subject placement, (iii) filled pauses, and (iv) coda /s/. Findings suggest modest but measurable intergenerational change for the lower-salience variables—such as subject pronoun expression, filled pauses, and subject placement—while higher-salience variables like coda /s/ exhibit patterns of relative stability. These results suggest that features lower in salience may shift more predictably in response to bilingual pressure, while more salient features are shaped by local social pressures and speaker ideologies.

The present dissertation deepens this line of inquiry by focusing on *liquid variation*—the realization of rhotics (/ɾ/, /r/) and the lateral /l/—a feature that is dialectally distinctive, perceptually salient, and prominent in folk ideologies of Caribbean Spanish. Crucially, speakers in this study also directly comment on liquid variation in metalinguistic discourse, underscoring its salience not only in broader ideological narratives but also in local speaker awareness. While liquid variation has been extensively studied in Spanish sociolinguistics, its relationship to salience in contact settings remains underexplored. By situating this variable alongside both high- and low-salience variables, this study probes whether liquid variation patterns more like coda /s/ or more like lower-salience variables. The goal is not to argue that highly salient variables resist contact-induced change, but rather that speakers are more likely to treat them independently, in ways shaped by local social pressures, ideological stances, and communicative goals. Unlike lower-salience variables—which often show convergence across generations in ways that reflect broad cognitive pressures—salient variables are less subject to such uniform pressures. Their social visibility makes them more available for stylistic use, but also renders their trajectories more variable across settings. By drawing together existing insights into salience and organizing them within a continuum-based framework, this dissertation offers an analytic approach to understanding how different types of variables are implicated in linguistic change, stability, and identity negotiation in contact settings.

1.3 Hypotheses and Contributions

Guided by the research questions, this dissertation hypothesizes the following patterns of linguistic variation:

LOWER-SALIENCE features (e.g., pronoun presence, subject placement, filled pauses) will exhibit significant convergence with English norms. Their lack of social visibility makes them more susceptible to so-called bilingual optimization strategies, allowing for structural realignment with minimal social cost.

HIGHER-SALIENCE features (e.g., coda /s/, liquid variation) will serve as resources for social meaning, with their realization shaped by speaker ideology and communicative goals. While these features may maintain alignment with national and regional dialectal norms, they are also subject to reconfiguration depending on whether speakers aim to reinforce dialectal differences, signal group membership, or accommodate external norms.

This research makes several contributions to sociolinguistic theory and methodology:

BROADENING THE SALIENCE FRAMEWORK- By incorporating liquids into salience studies, this dissertation refines the theoretical understanding of salience as a continuum rather than a binary property.

Expanding the Study of Co-variation- By analyzing both low- and high-salience variables together, this research adds to the growing research focusing of how linguistic features interact and shift in bilingual contexts.

Contextual Insights- Findings are situated within the unique sociolinguistic dynamics of the Boston community, emphasizing the role of social meaning, language ideologies, and identity in shaping linguistic behavior.

1.4 Structure of the Dissertation

Following the introduction, this dissertation is organized into six chapters, each addressing different aspects of the study:

Chapter 2- This chapter provides a comprehensive review of the literature, exploring salience, covariation, and the variables central to this study.

Chapter 3- This chapter details the methodology, including data collection procedures, acoustic analysis, and the role of social variables in the analysis. It also describes the *Spanish in Boston Corpus* and the sociolinguistic background of the community.

Chapter 4 - This chapter examines speaker profiles, highlighting their demographic backgrounds, patterns of language use, and attitudes toward dialectal variation. It lays the groundwork for understanding the sociolinguistic context of the participants' speech and provides critical insights into how their language use and attitudes interact with broader patterns of variation.

Chapter 5- This chapter presents an in-depth examination of liquid variation, focusing on the linguistic predictors and social factors that influence the realization of /ɾ/,

/r/, and /l/.

CHAPTER 6- This chapter explores potential covariation between liquids and other variables, situating the relationships between variables within a framework of salience and contact-sensitive patterns of variation.

CHAPTER 7- This chapter synthesizes findings from all previous chapters, discussing their implications for sociolinguistic theory, language contact, and the role of salience in bilingual communities. It also highlights the study's contributions and offers suggestions for future research directions.

1.5 Significance of the Study

By examining the interplay between salience, covariation, and language- and dialect-contact, this dissertation contributes to a more nuanced understanding of how bilingual speakers navigate complex sociolinguistic terrain. The findings show that not all variables respond similarly to contact pressures: some exhibit measurable shifts across generations, while others remain relatively stable, or display variability that defies straightforward patterns. This variability underscores an important point—contact-induced change is not inevitable, nor is it uniformly distributed across the linguistic system.

Rather than dividing variables into those that "converge" and those that are "maintained for social indexing," this study highlights how speaker awareness and social meaning shape the treatment of different variables. Variables that are lower in salience tend to pattern together, showing modest convergence consistent with bilingual optimization strategies. In contrast, more salient variables—those that are perceptually prominent, ideologically loaded, or explicitly referenced by speakers—are treated more independently. Their outcomes are more sensitive to local social dynamics and less constrained by cognitive pressures. In some communities, this may lead to maintenance or even amplification; in others, salient variables may undergo leveling, depending on how speakers orient to them. In doing so, the study builds on foundational insights from Trudgill (1986) and Labov (1972), offering further evidence that the degree to which speakers are aware of a variable—its salience—plays a key role in its behavior in contact settings. But importantly, this awareness does not make outcomes more predictable; rather, it opens the door to greater variability, since socially salient variables are more likely to be taken up as stylistic resources, sites of differentiation, or targets of reconfiguration.

By placing liquid variation within this broader framework, this dissertation offers a structured approach to understanding how salience and speaker agency shape the trajectories of linguistic variables in contact settings—showing that what changes, what remains stable, and why, depends not just on structure, but on social context, speaker goals, and community ideologies.

5 Results: An Examination of Liquids

This chapter delves into the variable production of Spanish liquids—the rhotics (/ɾ/) & /r/) and the lateral /l/—within a *comparative variationist framework* (Poplack & Levey 2010). This approach, widely applied in studies of Spanish in the U.S. (Otheguy & Zentella 2012; Erker 2017; Torres Cacoullos & Travis 2010; Erker & Otheguy 2016; Torres Cacoullos & Travis 2018), enables a systematic examination of variation in contact settings by comparing patterns of variability across measures of contact, such as immigrant generation or duration of life spent in a given locale. Building on the conceptual foundation laid in previous chapters, this analysis addresses key questions: What linguistic and social factors drive variation in the production of liquids? How do liquids align with the continuum of salience, and what does this reveal about their patterns of stability or change?

By "change," I refer to differential usage across speakers as a function of linguistic and social predictors, including Percentage of Life in the U.S. (PLUS), language use with interlocutors, and internal phonological constraints. If liquid variation aligns with patterns that have been observed among low-salience features, speakers with greater exposure to English—whether measured by PLUS, interactional patterns, or other contact-related measures—may exhibit variants more closely resembling English norms, reflecting the output of purported processes of bilingual optimization (Muysken 2013). Conversely, if liquid variation patterns with coda /s/, its use may function as a symbol of dialectal identity, resisting English-aligned variants despite contact. The potential for liquids to serve as an *identity shibboleth*—a marker of national pride or in-group membership—offers an important test of the relationship between social salience and linguistic behavior in contact settings.

To investigate these questions, I examine the distribution of liquid forms, assess their alignment with phonological expectations, and evaluate predictors of variation through statistical modeling. While the focus here is on identifying the linguistic and social factors that condition variation in liquids, the following chapter explores their covariation with other sociolinguistic variables, offering a broader perspective on coherence in linguistic systems.

5.1 General Overview of the Variation in the Dependent Variable

A total of 7,354 tokens of *liquids* were analyzed, with /ɾ/ (tap) having the highest number of tokens (n = 3,871), followed by /l/ (lateral, n = 2,626) and /r/ (trill, n = 857). As outlined in Chapter 3 (see Table 3.1), the phonemic distinctions and expected allophonic distributions for these liquids are based on standard accounts of Spanish phonology, particularly following Hualde (2014). Figure 5·1 below displays the proportional distribution of surface forms for each of the three liquids under study. In this context, *match* denotes surface forms that are identical to the phoneme, while *mismatch* refers to those that differ from the phonological form.

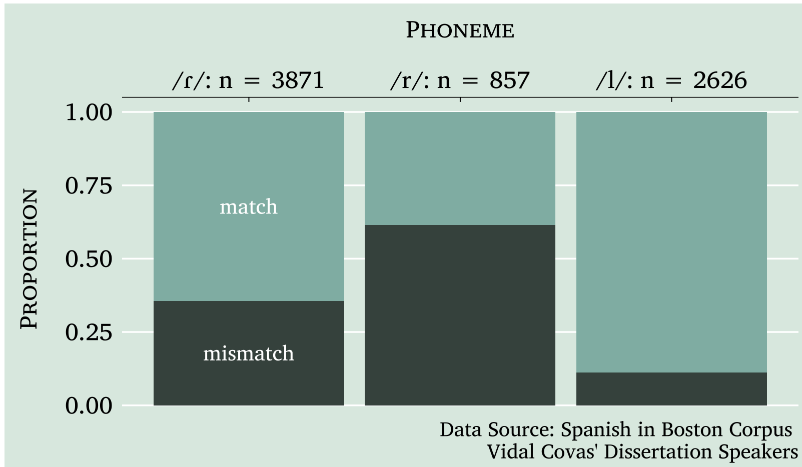

Figure 5·1: Rates of phonological and phonetic (mis)match for all liquids

Figure 5·2 expands upon Figure 5·1 by providing a detailed breakdown of distribution of the surface forms associated with each of the three phonemes: /ɾ/ (tap), /r/ (trill), and /l/ (lateral). The visualization illustrates the distribution of each surface form and categorizes them based on whether they align with the phonological form (*match*) or deviate from it (*mismatch*). The light shading indicates instances of *match*, highlighting surface forms that conform to the phonological expectations, while the dark shading represents *mismatch*, emphasizing phonetic realizations that are different from the phoneme.

Each panel corresponds to one of the three phonemes and reflects their relative token frequencies. Within each panel, surface forms are ordered by their frequency, revealing notable patterns in phonetic variation. For instance, the lateral /l/ demonstrates a high proportion of conformity to the canonical [l], with relatively few tokens exhibiting *mismatch*. In contrast, the tap /ɾ/ shows greater phonetic diversity, with [ɾ] as the dominant form but substantial representation of mismatched realizations such as [l], [ʔ], and [∅]. Similarly, the trill /r/ displays a high frequency of the mismatch form [r̞], which is even more common than the trill [r] itself, with the latter being the next most frequent surface form.

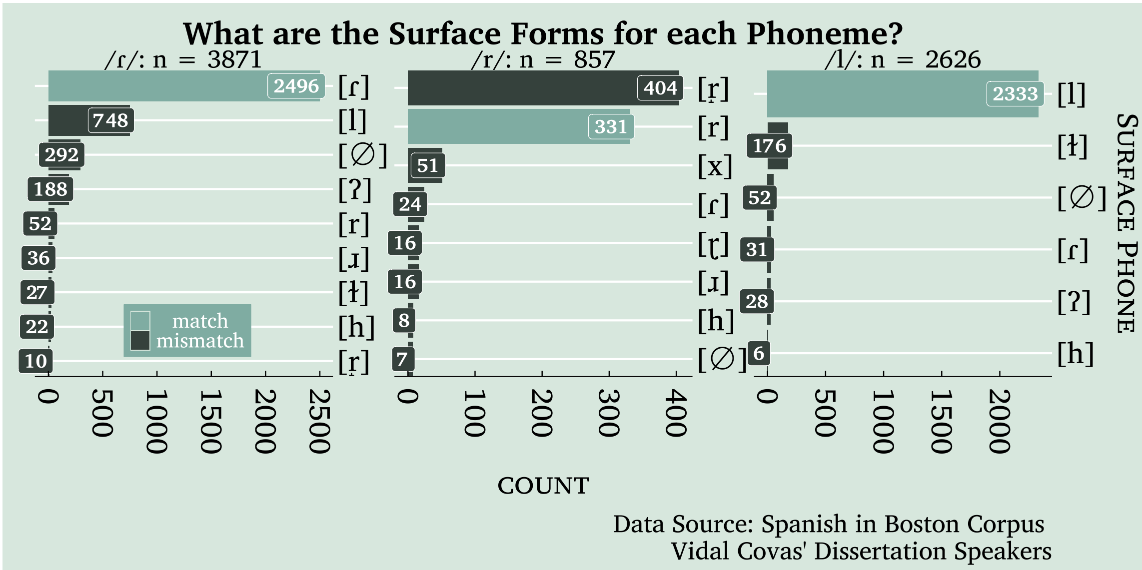

Figure 5·2: Rates of liquids

Having established the overall proportional distribution of surface forms, we now turn our attention to a more detailed examination of the three phonemes under study: /r/, /ɾ/, and /l/. In this analysis, I use the match/mismatch variable to categorize surface forms based on their correspondence with the purported phonological form. As previously defined, a *match* occurs when the surface form aligns with the phonological form, while a *mismatch* reflects a discrepancy between the two. This categorical distinction allows for the use of logistic regression, facilitating a structured statistical evaluation of the relationship between surface and phonological forms across the data set. Having outlined the model selection process in Section 3.3, we now turn to the detailed results of statistical modeling of variation for each phoneme.

5.2 Analysis of Tap, Trill, and Lateral

For each liquid, I present the best-fitting mixed-effects logistic regression model, identified using the `dredge()` and `drop1()` functions, and discuss key findings related to mismatch. The goal of these regressions is to estimate the probability of a mismatch as a function of several social and linguistic predictors, while accounting for both *fixed effects* (the linguistic and social predictors) and *random effects* (speaker). Including *speaker* as a random effect allows us to account for individual variation between speakers that might influence the outcome but are not directly related to the fixed effects. In other words, by incorporating random effects, we can partition the variability into that attributable to the speakers versus the predictors. For this analysis, I opted not to include *word* as an additional random effect because the primary focus is on inter-speaker variability rather than variability associated with specific lexical items. This decision balances model complexity with interpretability as well as model convergence likelihood, but future work could explore the potential influence of word-level variability. Relatedly, random slopes, which would allow the effect of specific predictors to vary by speaker, were not included in this analysis to avoid overparameterizing the models given the size of the dataset.

The r^2 values provide an indication of the model's explanatory power. The *marginal* r^2 reflects the proportion of variance explained by the fixed effects, while the *conditional* r^2 represents the variance explained by both the fixed and random effects (e.g., by-speaker variability). The *Intraclass Correlation Coefficient (ICC)* measures the proportion of total variance in the outcome attributable to differences between speakers, as opposed to variability within speakers (i.e., between observations from the same speaker).

In each model summary, the β *(estimate)* represents the change in the odds of a mismatch associated with a one-unit increase in a continuous predictor variable or, in the case of a categorical variable, the change relative to the reference level, holding all other variables constant. It is important to note that β is not a direct measure of effect size; rather, its magnitude reflects the relationship between the predictor and the odds of the outcome. A positive estimate indicates that the predictor increases the likelihood of a mismatch, while a negative estimate indicates a decrease. The *standard error (SE)* reflects the variability of the estimate due to sampling, while the *confidence interval (CI)* provides a range within which the true value of the parameter is likely to fall, given a specified level of confidence (e.g., 95%). The *z-statistic* reflects the standardized distance of the estimate from zero, and is used to calculate the *p-value*, which indicates whether the predictor's effect is statistically significant.

5.2.1 Statistical Analysis of Variation in the Tap — /ɾ/

The best-fitting model for tap variation, as shown in Table 5.1, includes the predictors *country of origin*, *following sound category*, *log lexical frequency*, *milliseconds per syllable*, *PLUS (percent life in the US)*, *position in syllable*, *sex*, and *stress*. It is important to note that inclusion in the best-fitting model does not necessarily imply that a variable is a significant predictor; for instance, *PLUS* exhibits a weak effect and is not statistically significant in this model (see Table 5.1). However, as demonstrated in Figure 5.4, even non-significant predictors can still contribute to the overall fit of the model. This figure visually illustrates the impact of each predictor on model AIC, highlighting that removing *PLUS* results in a small but measurable change in model fit, which justifies its retention in the analysis.

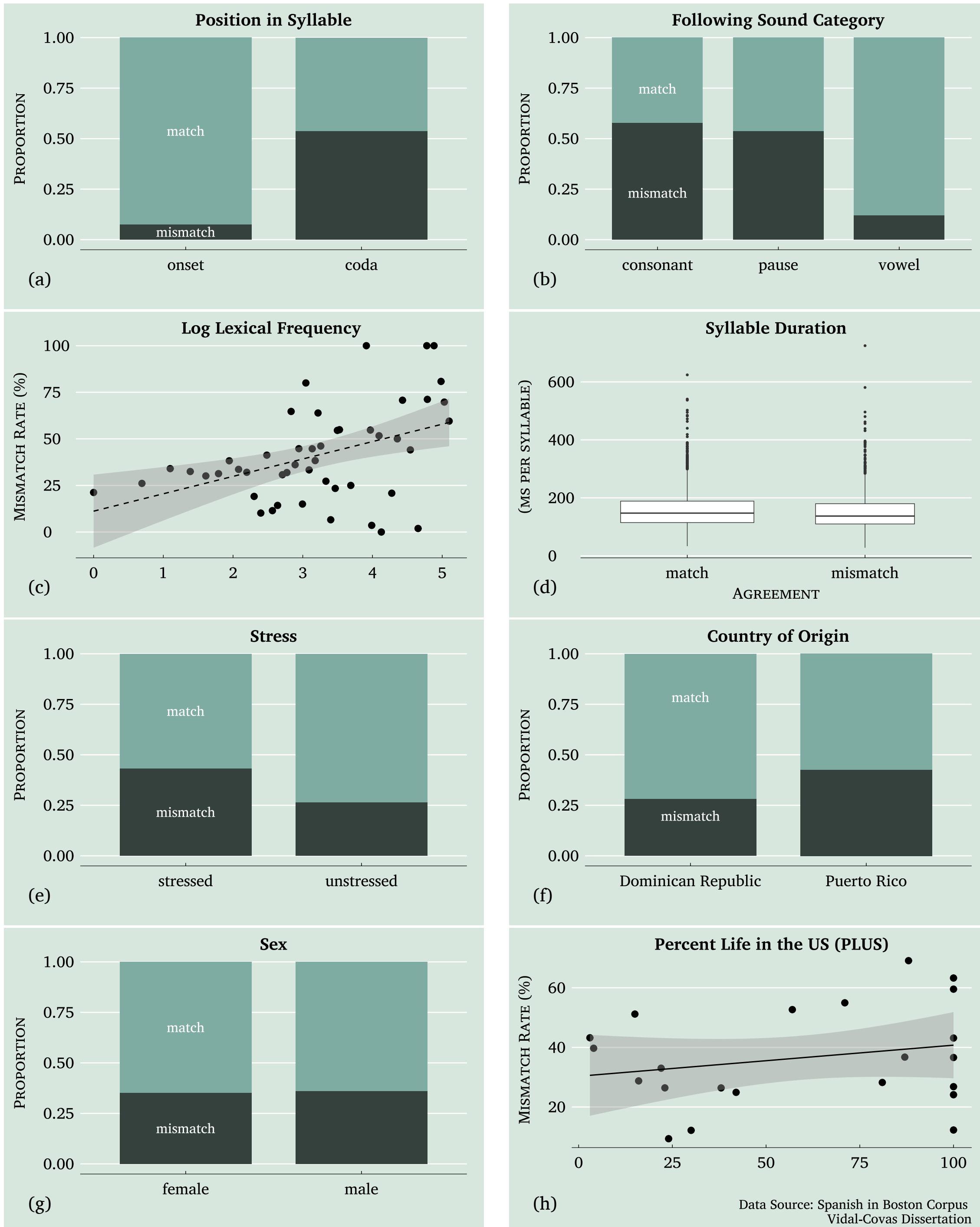

Figure 5·3: Visual representation of the effects of the predictors included in the best-fitting mixed effects model for tap mismatch

Table 5.1: Mixed Effects Model: Tap Mis-(match) Prediction

	Dep. Variable: Mismatch				
Predictors	β	SE	CI 95 %	z-stat	p-value
Intercept	-3.25	.51	[-4.24−-2.26]	-6.42	.000***
Country of Origin *(ref:DR)*					
Puerto Rico	1.03	.39	[.27−1.79]	2.65	.008***
Following Sound Category *(ref:cons)*					
Pause	-.35	.17	[-.68−-.03]	-2.13	.033**
Vowel	-1.51	.14	[-1.79−-1.24]	-10.67	.000***
Log Lexical Frequency	.16	.03	[.11−.22]	5.70	.000***
Milliseconds per Syllable	-.22	.05	[-.32−-.12]	-4.29	.000***
PLUS	.01	.01	[-.00−.02]	1.56	.119
Position in Syllable *(ref:onset)*					
Coda	2.05	.18	[1.70−2.41]	11.21	.000***
Sex *(ref:female)*					
Male	.85	.39	[.09−1.61]	2.20	.028**
Stress *(ref:stressed)*					
Unstressed	-.43	.10	[-.62−-.23]	-4.29	.000***
Random Effects					
σ^2	3.29				
τ_{00} Speakers	.77				
ICC	.19				
N Speakers	22				
SD Speakers	.88				
Observations	3871				
Marginal R^2/ Conditional R^2	.46 / .56				
AIC	3364.36				

$^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$

The fixed effects in the model reveal several significant results. ***Position in syllable: coda*** ($\beta = 2.05$, $p < .001$) significantly increases the likelihood of a mismatch, indicating that codas are associated with a higher probability of mismatch. Likewise, ***following sound category: vowel*** ($\beta = -1.51$, $p < .001$) and ***log lexical frequency*** ($\beta = .16$, $p < .001$) demonstrate significant effects, with vowels following taps reducing mismatch likelihood, while increased lexical frequency raises it. These relationships are visually represented in Figure 5·3, which illustrates the distribution of mismatch patterns across the different predictors. Specifically, Subfigure 5·3a highlights the increased likelihood of mismatches in coda positions, and Subfigure 5·3b shows the distribution of phonetic (mis)match across following sound categories.

To assess the impact of each predictor on model fit, I examined the increase in *Akaike information criterion (AIC)* if each predictor were removed from the model, summarized in Figure 5·4. The largest increases in AIC were observed for *position in syllable* (129.8), *following sound category* (118.3), and *log lexical frequency* (30.9), indicating that these predictors contribute substantially to the model's explanatory power. Conversely, *PLUS* showed a negligible AIC increase (0.3) and a non-significant p-value, suggesting its limited effect on model fit.

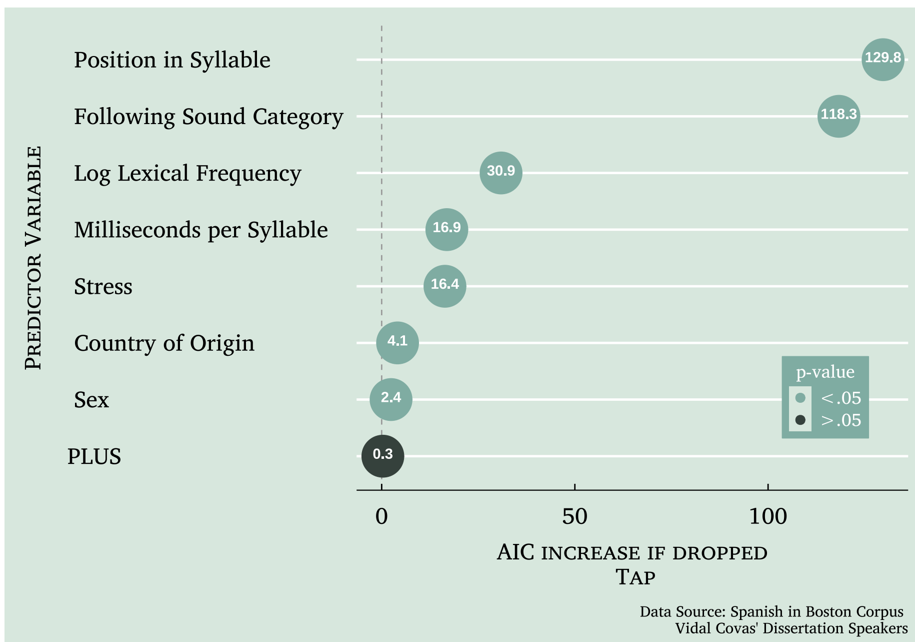

Figure 5·4: Impact of predictor variable removal on AIC in mixed effects model — /ɾ/

Conversely, *milliseconds per syllable* ($\beta = -.22$, $p < .001$) and *stress: unstressed* ($\beta = -.43$, $p < .001$) have negative regression coefficients, indicating that longer syllable durations and unstressed syllables are associated with a decreased likelihood of a mismatch. Although *PLUS* ($\beta = .01$, $p = .119$) is not statistically significant, it was retained in the model because removing it would increase the AIC, indicating a reduction in overall model fit. Subfigure 5·3d shows that longer syllable durations correspond to fewer mismatches, and Subfigure 5·3e shows that unstressed syllables are more prone to mismatches compared to stressed syllables.

The random effects for *speaker* show a variance of $\tau_{00} = 0.77$, with the *ICC* of $.19$, indicating that 19% of the variance in mismatch is due to differences between speakers. The model's *marginal r*^{2} is $.46$, representing the variance explained by fixed effects alone, while the *conditional r*^{2} is $.56$, showing that the full model (including random effects) explains 56% of the variance.

5.2.1.1 Summary of Findings for Tap Variation

The results from the statistical modeling of tap variation reveal key patterns in how mismatches occur and the factors that influence them.

First, *position in syllable* plays a crucial role: taps occurring in *coda* position are much more likely to surface as a mismatch than those in onset position. This is not surprising and aligns with previous findings on segmental weakening in Spanish, where coda consonants are more susceptible to phonetic reduction or modification (Poplack 1979; Alba 1990; Quilis 1993; Ramos-Pellicia 2004; Beaton 2016; Gilbert & Rohena-Madrazo 2017, inter alia). Similarly, *following sound category* has a strong effect. Taps followed by *vowels* are much more likely to match their phonological target, while those followed by *consonants* or *pauses* have a higher probability of mismatch. This suggests that coarticulatory effects and articulatory constraints influence realization patterns, with vowel-adjacent taps benefiting from greater stability. Lexical factors also shape variation: *higher-frequency words* show an increased likelihood of mismatch.

Temporal factors are also relevant. The results show that *longer syllable durations* reduce mismatch likelihood. This suggests that taps produced in *faster speech* are more prone to non-canonical realizations, reinforcing the idea that mismatch rates may be linked to speech rate and articulatory effort. This pattern aligns with claims by Katz (Katz 2021), who argues that many lenition effects may be explained in terms of duration. The effects of *stress* further support this trend. *Unstressed syllables* are significantly more likely to show matches than stressed ones. This pattern, however, reflects a tendency *within the Spanish tap data presented here*, rather than a general phonological trend. In fact, it runs counter to common cross-linguistic patterns, where reduction is typically favored in prosodically weaker positions. The fact that, in this dataset, Spanish taps show increased reduction in prosodically stronger positions (i.e., stressed syllables) highlights a point of typological interest and suggests that tap realization in this variety of Spanish may diverge from more general lenition patterns — a finding that warrants further investigation.

Sociolinguistic factors also emerge as significant predictors. Speakers from *Puerto Rico* exhibit higher mismatch rates compared to those from the *Dominican Republic*, suggesting dialectal differences in the realization of the tap. Additionally, *male speakers* show higher rates of mismatch than female speakers, aligning with prior sociophonetic research on gender-based variation in liquid reduction and articulation (Beaton 2005). Finally, the effect of *PLUS* was weak and not statistically significant. However, it was retained in the model because removing it increased AIC, indicating that—even if not individually significant—it still contributed to model fit in some capacity.

5.2.2 Tap Subset: Laterals Mismatch

To better understand the factors contributing to tap variation, I conducted a subset analysis focusing on lateral mismatches—the most frequent type of mismatch observed in the dataset. The full dataset comprised 3,871 tap tokens, of which 3,244 were included in this subset. Within the subset, 2,496 tokens were realized as [ɾ], making up 76.94% of the data. The most common mismatch, [l], was produced in 748 cases, occurring in 23.06% of the subset. Given its relatively high occurrence and its classic association with so-called *liquid neutralization*, this subset analysis aimed to identify the linguistic and social factors that most strongly influenced this particular realization. Table 5.2 presents the optimal model for predicting lateral mismatches for tap.

Table 5.2: Mixed Effects Model: Tap Match vs. Lateral Mismatch Prediction

	Dep. Variable: Mismatch				
Predictors	β	SE	CI 95 %	z-stat	p-value
Intercept	-6.05	.88	[-7.77–-4.33]	-6.88	.000***
Country of Origin *(ref:DR)*					
Puerto Rico	1.83	.51	[.84–2.82]	3.62	.000***
Following Sound Category *(ref:cons)*					
Pause	-.31	.20	[-.71–.09]	-1.50	.132
Vowel	-2.11	.20	[-2.50–-1.72]	-10.60	.000***
Log Lexical Frequency	.13	.04	[.06–.20]	3.43	.001***
Milliseconds per Syllable	-.19	.07	[-.32–-.05]	-2.74	.006***
Position in Syllable *(ref:onset)*					
Coda	2.55	.34	[1.88–3.22]	7.45	.000***
Vowel	1.74	.76	[.24–3.23]	2.28	.023**
Sex *(ref:female)*					
Male	1.21	.51	[.22–2.20]	2.39	.017**
Stress *(ref:stressed)*					
Unstressed	-.82	.13	[-1.08–-.55]	-6.06	.000***
Random Effects					
σ^2	3.29				
$\tau_{00\ \text{Speakers}}$	1.29				
ICC	.28				
N $_{\text{Speakers}}$	22				
SD $_{\text{Speakers}}$	1.13				
Observations	3244				
Marginal R^2/ Conditional R^2	.66 / .75				
AIC	1955.22				

$^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$

The predictors included in this model are *country of origin*, *following sound category*, *log lexical frequency of word*, *milliseconds per syllable*, *position in syllable*, *previous sound category*, *sex*, and *stress*. Model selection was conducted using the `dredge()` function, which identified the best-fitting model based on AIC. Additionally, the `drop1()` function confirmed that all retained predictors contributed meaningfully to model fit, as removing any one of them resulted in an increase in AIC.

The random effects structure indicates substantial between-speaker variability, with the *ICC* at $.28$, suggesting that nearly a third of the variation stems from differences across speakers. The *marginal* r^2 of 0.66 highlights the significant explanatory power of fixed effects alone, while the *conditional* r^2 of $.75$ underscores the role of both fixed and random effects.

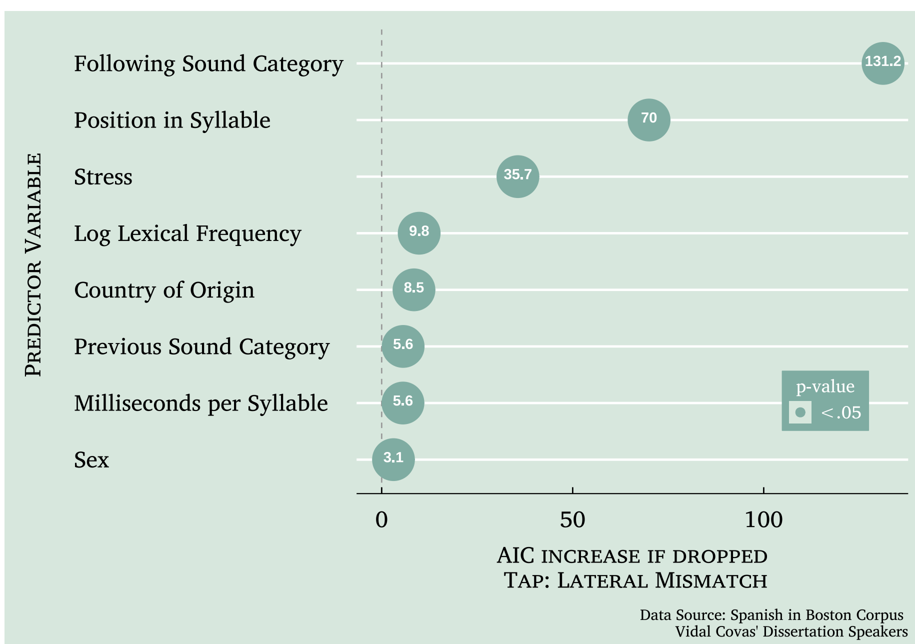

Figure 5.5: Impact of predictor variable removal on AIC in mixed effects model — /ɾ/: [ɾ] & [l]

The relative impact of each predictor on the model's AIC is further visualized in Figure 5.5, which provides a clear representation of how dropping each variable affects the overall model fit. In this model, *following sound category* proved to be the most influential predictor, as shown by its large AIC increase (131.2) if dropped. This suggests that the phonetic context following a tap plays a critical role, particularly in determining the likelihood of surfacing as a lateral form. Position in Syllable, also significant in the broader tap mismatch model, exerts a strong influence here, with coda positions strongly favoring lateral mismatch, though its effect is secondary to Following Sound Category.

The effect of *stress* is also significant, with unstressed syllables less prone to lateral mismatches—a pattern consistent with findings in the broader tap mismatch analysis. Although predictors like *log lexical frequency* and *milliseconds per syllable* are less impactful, they still contribute meaningfully, highlighting the relevance of lexical and temporal dynamics in mismatch occurrences.

In both the overall tap mismatch model and the lateral subset, two social factors—*country of origin* and *sex*—stand out as significant contributors. In the subset analysis, Puerto Rican speakers and males are more prone to lateral mismatches, with these social factors ranking higher in their influence on AIC than some linguistic factors, such as *previous sound category* and *milliseconds per syllable*. This reinforces the importance of social variables in explaining variation particularly for this subset of the data, adding depth to the analysis beyond the purely linguistic factors.

Interestingly, *previous sound category*, included in this subset model, provides additional insights by emphasizing the impact of the preceding phonetic environment. Vowel sounds before the tap significantly predict lateral mismatch, highlighting a nuanced aspect of phonetic context for this specific surface form. Finally, unlike in the overall tap mismatch model, the *PLUS* predictor was excluded here, as it did not significantly affect the subset model's fit.

5.2.2.1 Summary of the Results for Tap Subset: Laterals Mismatch

In terms of the model examining lateral mismatches for taps, the strongest predictor in the model was *following sound category*, indicating that the segment following the tap significantly affects whether it surfaces as a lateral. Specifically, mismatches were most likely when the following sound was a consonant, while vowels and pauses significantly reduced mismatch probability.

Additionally, *position in syllable* was a key factor, with coda positions favoring lateral mismatches. Similarly, *stress* influenced mismatch likelihood, with unstressed syllables less prone to lateralization. While less influential than in the larger analyses of tap, *log lexical frequency* and *milliseconds per syllable* also contributed to the model analyzing lateral realizations of tap, suggesting that more frequent words and faster speech rates may slightly influence lateralization patterns. Interestingly, *previous sound category* was included as a predictor, revealing that lateral mismatches were more likely when the preceding sound was a vowel.

Social factors also played an important role in lateral mismatches. *Country of origin* was a significant predictor, showing that Puerto Rican speakers exhibited more lateral mismatches than Dominican speakers. Likewise, *sex* had a measurable effect, with male speakers more likely to produce lateral mismatches. These findings highlight the importance of incorporating social factors, which contribute to variability in tap realization beyond purely linguistic conditioning.

5.2.3 Statistical Analysis of Variation in the Trill — /r/

The best-fitting model for the trill, shown in Table 5.3, includes the predictors *country of origin*, *log lexical frequency*, *milliseconds per syllable*, *PLUS (percent life in the US)*, and *position in word*.

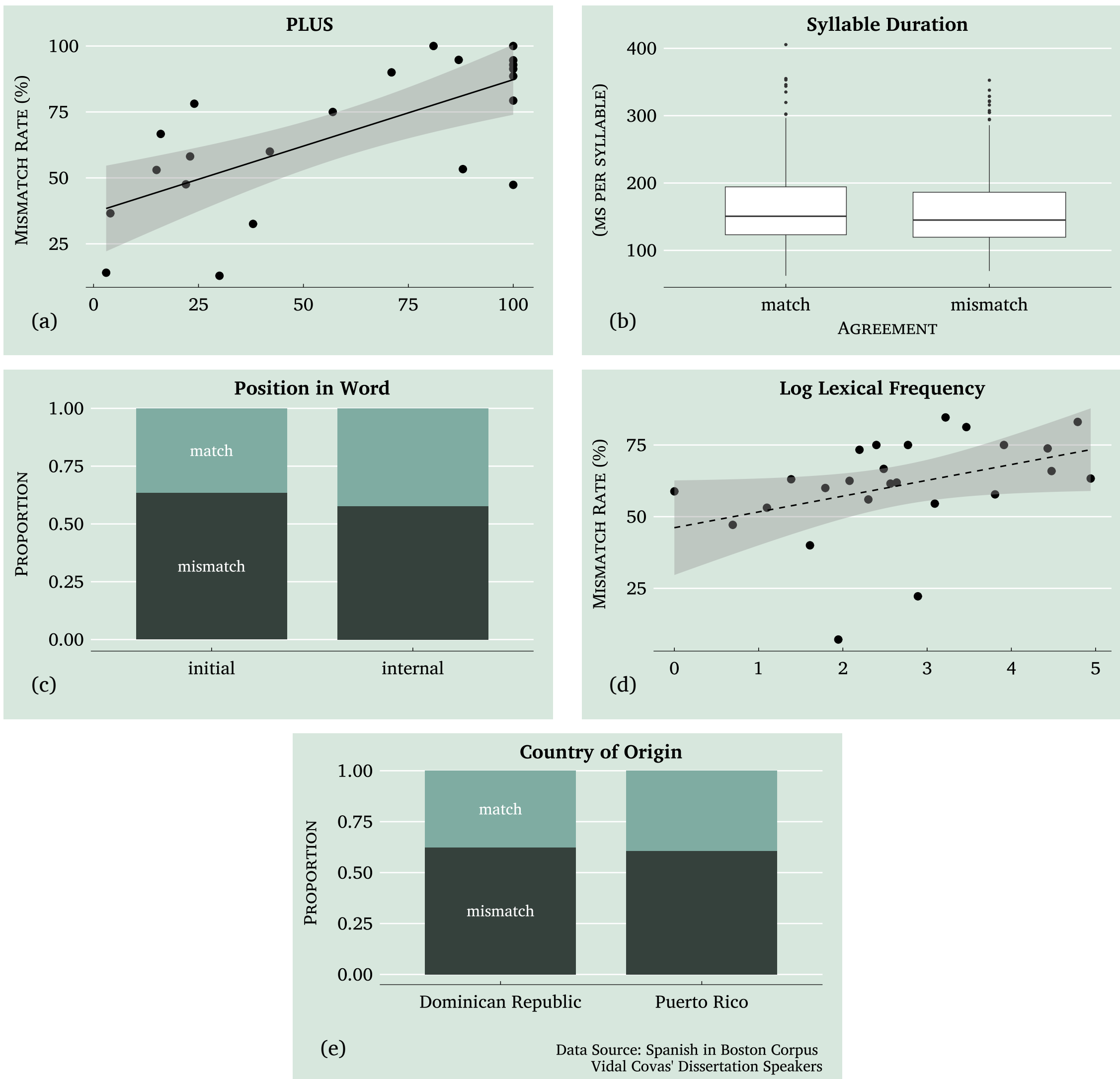

Figure 5·6: Visual representation of the effects of the predictors included in the best-fitting mixed effects model for trill mismatch

The fixed effects in the model reveal several significant effects. Higher *log lexical fre-*

quency ($\beta = .15$, $p = .006$) significantly increases the likelihood of a mismatch, suggesting that as lexical frequency rises, so does the probability of mismatch. Similarly, *PLUS* ($\beta = .03$, $p < .001$) shows a positive relationship with mismatch probability, suggesting that a higher percentage of life spent in the US is associated with more frequent mismatches. This relationship is visually represented in Subfigure 5·6a.

Table 5.3: Mixed Effects Model: Trill Mis-(match) Prediction

	Dep. Variable: Mismatch				
PREDICTORS	β	SE	CI 95 %	Z-STAT	P-VALUE
Intercept	-.74	.51	[-1.73−.26]	-1.44	.149
Country of Origin *(ref:DR)*					
Puerto Rico	-.81	.51	[-1.80−.19]	-1.59	.112
Log Lexical Frequency	.15	.05	[.04−.25]	2.76	$.006^{***}$
Milliseconds per Syllable	-.30	.09	[-.48−-.13]	-3.39	$.001^{***}$
PLUS	.03	.01	[.02−.05]	4.81	$.000^{***}$
Position in Word *(ref:initial)*					
Internal	-.57	.18	[-.93−-.22]	-3.15	$.002^{***}$
RANDOM EFFECTS					
σ^2	3.29				
$\tau_{00\ \text{SPEAKERS}}$	1.07				
ICC	.25				
N_{SPEAKERS}	22				
SD_{SPEAKERS}	1.03				
Observations	857				
Marginal R^2/ Conditional R^2	.28 / .45				
AIC	907.35				

$^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$

Conversely, *milliseconds per syllable* ($\beta = -.30$, $p = .001$) and *position in word: internal* ($\beta = -.57$, $p = .002$) have negative regression coefficients, indicating that slower speech and internal word positions reduce the likelihood of a mismatch. Subfigures 5·6b and 5·6c depict these effects, illustrating how mismatches are less likely in internal word positions and with longer syllable durations. Additionally, *country of origin: PR* has a negative but non-significant effect ($\beta = -0.81$, $p = .112$), implying fewer mismatches among Puerto Rican speakers. Although this effect is not statistically significant, it is included to maximize model fit, as demonstrated in Subfigure 5·6e, which shows patterns of phonetic/phonological (mis)match by country.

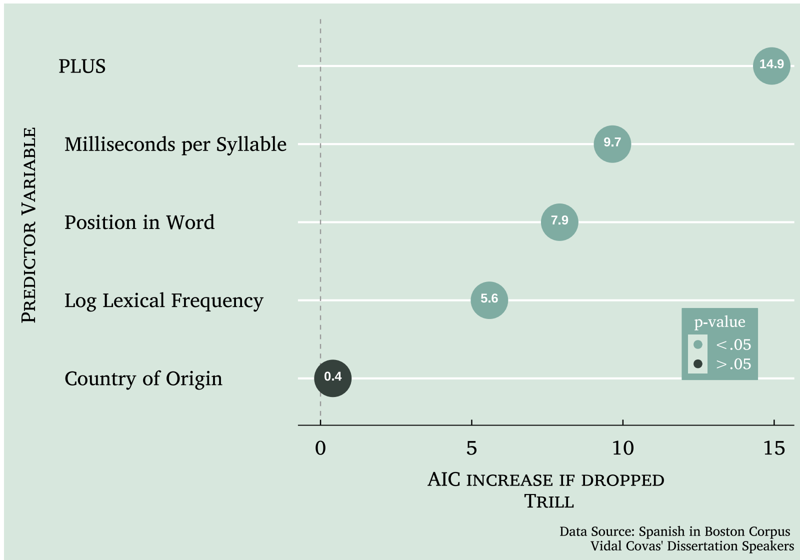

Figure 5·7: Impact of predictor variable removal on AIC in mixed effects model — /r/

The random effects for *speaker* show a variance of $\tau_{00} = 1.07$, and the *ICC* is $.25$, indicating that 25% of the total variance in mismatch is attributable to differences between speakers. The model's explanatory power is reflected in the *marginal* r^2 of $.28$, representing the variance explained by the fixed effects, and a *conditional* r^2 of $.45$, indicating that the full model (including random effects) explains 45% of the variance.

As shown in Figure 5·7, subsequent analysis using the `drop1()` function confirmed that all significant predictors in the final model contributed meaningfully to explaining the variation in mismatch outcomes. Although *country of origin* was not a statistically significant predictor ($p = .112$), its inclusion is justified because removing it from the model results in a marginal increase in AIC (.4 points), suggesting that its presence, while not contributing significantly in isolation, helps maintain the overall model fit.

5.2.3.1 Summary of the Results for Trill

The findings from this model highlight key patterns in trill mismatch variation, emphasizing the role of both linguistic and social factors in shaping this phonetic outcome. The significance of *log lexical frequency* suggests that trills in more frequently used words are more prone to mismatch. The relationship between *PLUS* and mismatch probability indicates that speakers who have spent a greater proportion of their lives in the U.S. are more likely to exhibit trill mismatches.

Phonetic factors such as *milliseconds per syllable* and *position in word* appear to reduce mismatch likelihood. The association between *milliseconds per syllable* and mismatch probability indicates that slower speech, characterized by longer syllable durations, corresponds with more consistent trill productions. Similarly, trills in *internal word positions* are less likely to undergo mismatch.

Although *country of origin* was not statistically significant, its inclusion in the model suggests some degree of differentiation between Puerto Rican and Dominican speakers in their realization of the trill.

5.2.4 Trill Subset: [r̝] Mismatch

To better understand the factors influencing trill variation, I conducted a subset analysis focusing on the most frequent mismatch type, [r̝]. The full dataset, as discussed above, contained 857 trill tokens, of which 735 were included in this subset. Within this subset, the match category—meaning tokens realized as [r]—accounted for 331 tokens or 45.03%, while the most common mismatch, [r̝], occurred in 404 tokens, representing 54.97% of the subset. Given that this mismatch type was the predominant realization, this analysis aimed to identify the linguistic and social factors that contributed to its occurrence. Key predictors in this model are *country of origin*, *log lexical frequency*, *milliseconds per syllable*, *PLUS (percent life in the US)*, *position in word*, and *sex*.

Table 5.4: Mixed Effects Model: /r/ Match vs. [r̝] Mismatch Prediction

	Dep. Variable: Mismatch				
Predictors	β	SE	CI 95 %	z-stat	p-value
Intercept	-.07	.53	[-1.10−.97]	-.12	.901
Country of Origin *(ref:DR)*					
Puerto Rico	-.90	.46	[-1.81−.01]	-1.95	.052*
Log Lexical Frequency	.14	.05	[.04−.25]	2.63	.009***
Milliseconds per Syllable	-.35	.09	[-.53−-.16]	-3.70	.000***
PLUS	.02	.01	[.01−.04]	3.85	.000***
Position in Word *(ref:initial)*					
Internal	-.72	.19	[-1.09−-.34]	-3.72	.000***
Sex *(ref:female)*					
Male	-.87	.46	[-1.77−.02]	-1.91	.056*
Random Effects					
σ^2	3.29				
τ_{00} Speakers	.83				
ICC	.20				
N Speakers	22				
SD Speakers	.91				
Observations	735				
Marginal R^2/ Conditional R^2	.24 / .39				
AIC	834.54				

$^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$

Notably, *milliseconds per syllable* and *position in word* emerged as the strongest internal predictors, with significant negative effects ($\beta = -.35$, $p < .001$ and $\beta = -.72$, $p < .001$, respectively). This suggests that longer syllable durations and internal word positions substantially reduce the likelihood of a mismatch to [r̝]. Moreover, *PLUS* had a significant positive effect ($\beta = .02$, $p < .001$), indicating more time in the US leads to a higher probability of /r/ surfacing as [r̝].

Conversely, other predictors, such as *log lexical frequency*, *country of origin: PR*, and *sex:male* were moderately significant ($\beta = .14$, $p < .05$, $\beta = -.90$, $p < .1$, and $\beta = -.87$, $p < .1$ respectively). These results indicate that mismatches are more likely when the word has a higher lexical frequency, in the speech of Dominican speakers compared to Puerto Rican speakers, and in the speech of females compared to males.

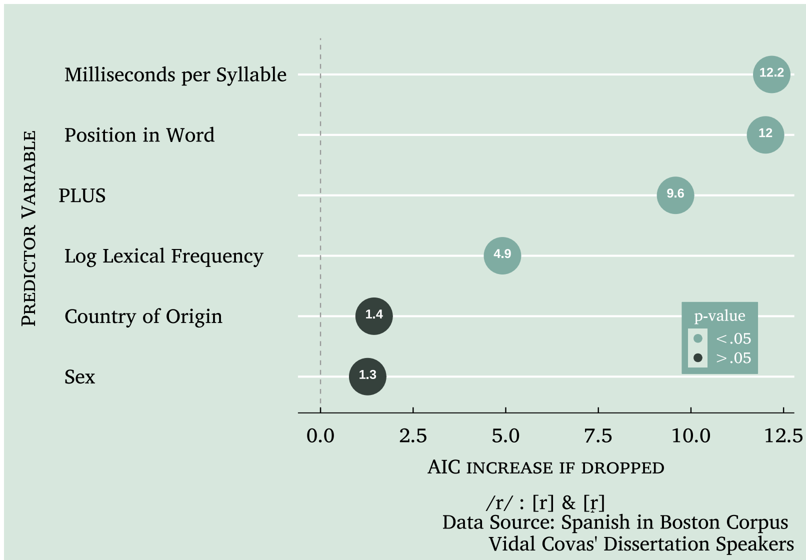

Figure 5·8: Impact of predictor variable removal on AIC in mixed effects model — /r/: [r] & [ɹ̠]

Using the `drop1()` function, I evaluated the impact of each predictor on the model's fit, as visualized in Figure 5·8. The most influential predictors were *milliseconds per syllable*, *position in word* and *PLUS*, each resulting in a notable increase in AIC (12.3, 12.1, and 9.6, respectively) when excluded.

The model explains a moderate portion of the variance, with a *marginal* r^2 of .24 for the fixed effects and a *conditional* r^2 of .39, including random effects. This indicates that the fixed effects alone account for 24% of the variance in mismatch outcomes, and the full model captures 39% when speaker-level variability is considered.

5.2.4.1 Summary of the Results for Trill Subset: [ɾ̞] Mismatch

The results of the trill subset analysis reveal several key patterns in the production of [ɾ̞] as a realization of trill. Among the strongest predictors, *milliseconds per syllable*, *position in word*, and *PLUS* are highly significant, with longer syllable durations (i.e., slower speech rates), internal word positions, and a lower PLUS associated with a lower likelihood of a trill being realized as [ɾ̞]. This suggests that faster speech, word-initial positions, and more time in the US favor the production of this particular mismatch.

Additionally, *log lexical frequency* was associated with a higher likelihood of the mismatch, suggesting that more frequently used words are more likely to surface as [ɾ̞]. *Country of origin* and *sex* were moderately significant predictors, with Puerto Rican and male speakers producing fewer instances of the variant than Dominican and female speakers.

5.2.5 Statistical Analysis of Variation in the Lateral — /l/

For the lateral liquid subtype, the best-fitting model, displayed in Table 5.5, includes the predictors *following sound category*, *stress*, *syllable type*, *% English only with interlocutors*, and *previous sound category*. As shown in Figure 5·10, *following sound category* emerged as the most influential predictor, with an AIC increase of 27.9 if omitted. Other significant predictors, such as *stress* (9.8) and *syllable type* (9.6), also show considerable impacts on the model's fit. By contrast, *previous sound category* contributed minimally, with an AIC increase of only 1.5, suggesting a limited role in explaining mismatch variability.

The fixed effects analysis highlighted that *following sound category: pause* ($\beta = -0.83$, $p = .007$) and *following sound category: vowel* ($\beta = -1.44$, $p < .001$) both significantly decrease the likelihood of a mismatch compared to consonants, indicating that mismatches

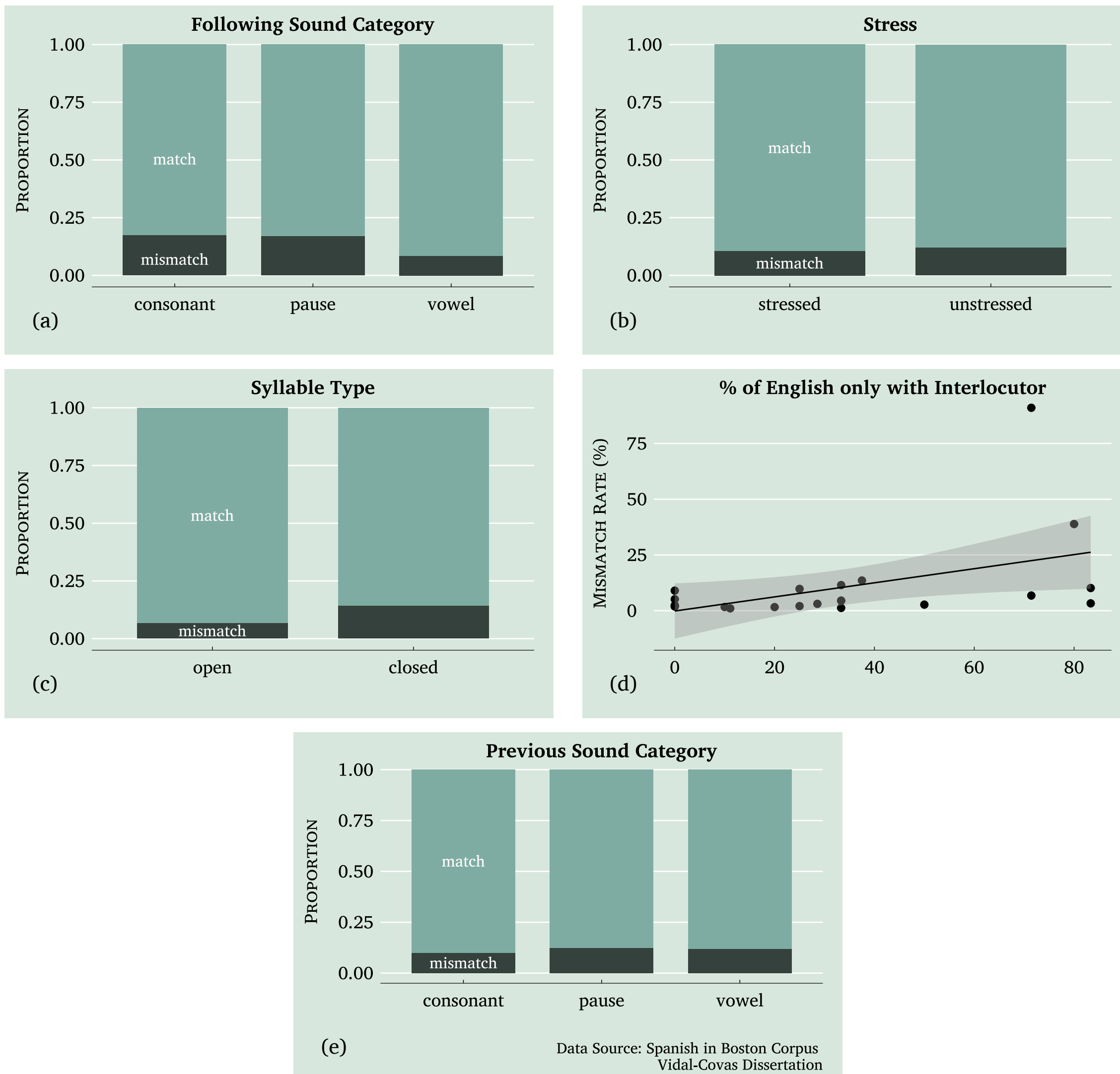

Figure 5·9: Visual representation of the effects of the predictors included in the best-fitting mixed effects model for lateral mismatch are more probable when the following sound is a consonant. Additionally, ***stress: unstressed*** ($\beta = 0.65$, $p < .001$) and ***syllable type: closed*** ($\beta = 0.83$, $p = .001$) have positive regression coefficients, suggesting that mismatches are more frequent in unstressed and closed syllables. The ***% English only with interlocutors*** ($\beta = 0.03$, $p = .004$) also showed a positive effect, indicating that higher percentages of English-only interactions increase mismatch likelihood.

Table 5.5: Mixed Effects Model: Lateral Mis-(match) Prediction

	Dep. Variable: Mismatch				
Predictors	β	SE	CI 95 %	z-stat	p-value
Intercept	-3.73	.63	[-4.96−-2.50]	-5.93	.000***
Following Sound Category *(ref:cons)*					
Pause	-.83	.31	[-1.43−-.23]	-2.72	.007***
Vowel	-1.44	.27	[-1.97−-.92]	-5.38	.000***
% English Only w Interlocutors	.03	.01	[.01−.05]	2.90	.004***
Previous Sound Category *(ref:cons)*					
Pause	-.86	.43	[-1.69−-.03]	-2.02	.043**
Vowel	-.38	.24	[-.86−.09]	-1.58	.115
Stress *(ref:stressed)*					
Unstressed	.65	.19	[.28−1.01]	3.49	.000***
Syllable Type *(ref:open)*					
Closed	.83	.24	[.36−1.30]	3.44	.001***
Random Effects					
σ^2	3.29				
$\tau_{00\ \text{Speakers}}$	1.88				
ICC	.36				
N Speakers	22				
SD Speakers	1.37				
Observations	2626				
Marginal R^2/ Conditional R^2	.23 / .51				
AIC	1104.61				

$^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$

Visual representations of these effects, shown in Figure 5·9, further illustrate the relationships between each predictor and mismatch rates. For instance, they show that mismatch likelihood is notably reduced in contexts where pauses or vowels follow the lateral. Unstressed syllables, conversely, are more likely to result in mismatches compared to stressed ones, and closed syllables tend to exhibit more mismatches than open syllables.

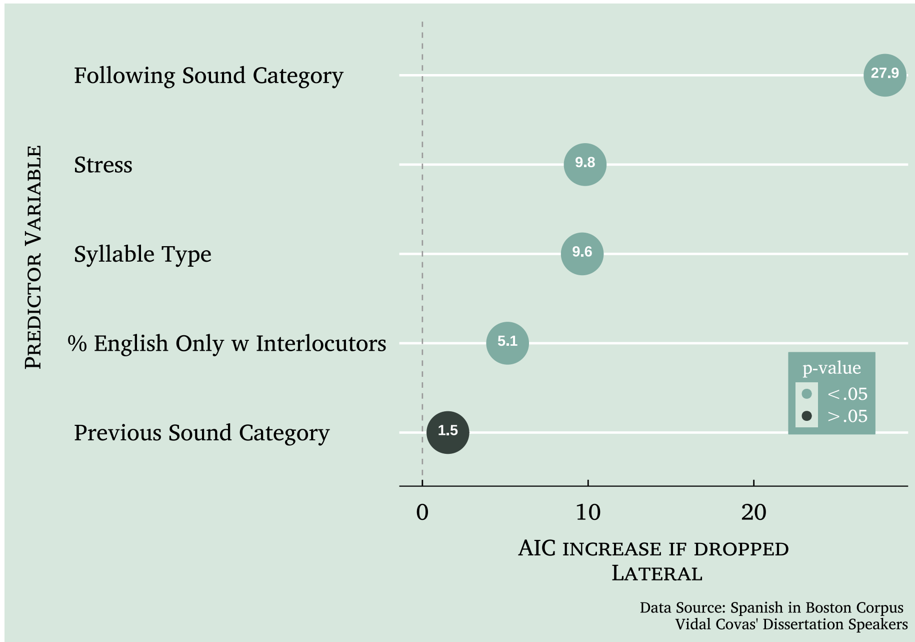

Figure 5·10: Impact of predictor variable removal on AIC in mixed effects model — /l/

The random effects variance for *speaker* is $\tau_{00} = 1.88$, with an *ICC* of $.36$, indicating that 36% of the variance in mismatches is due to differences between individual speakers. The *marginal r*^{2} of $.23$ reflects the variance explained by fixed effects alone, while the *conditional r*^{2} of $.51$ suggests that the full model accounts for 51% of the variance in mismatch outcomes.

5.2.5.1 Summary of the Results for Lateral

Among the most significant predictors, *following sound category* played a major role, with mismatches more likely when the following segment was a consonant, while vowels and pauses were associated with a reduced likelihood of mismatch.

Phonological factors such as *stress* and *syllable type* were also relevant. The positive regression coefficient for *stress: unstressed* indicates that laterals occurring in unstressed syllables were more prone to mismatch. Similarly, the effect of *syllable type: closed* suggests that closed syllables foster more frequent lateral mismatches than open syllables, potentially due to increased articulatory constraints in coda positions.

Social factors further contributed to variation in lateral mismatch. The significant effect of *% English only with interlocutors* indicates that speakers who engage in more English-only interactions exhibit a higher likelihood of mismatch. Additionally, while *previous sound category* was included in the model, its overall effect was minor, with mismatches slightly more likely after a consonant than after vowels or pauses.

5.2.5.2 Lateral Subset: Dark-L Mismatch

To better understand the factors influencing lateral variation, I conducted a subset analysis focusing on the most frequent mismatch type, [ɫ]. The full dataset, as discussed above, contained 2,626 lateral tokens, of which 2,509 were included in this subset. Within this subset, the match category—meaning tokens realized as [l]—accounted for 2,333 tokens, making up 92.99% of the data. In contrast, the most common mismatch, [ɫ], was found in 176 tokens, representing 7.01% of the subset. Given that this mismatch type was the predominant realization among non-matching tokens, this analysis aimed to identify the linguistic and social factors that contributed to its occurrence.

The model for this subset, shown in Table 5.6, examines the variation between non-velarized lateral [l] matches and velarized lateral [ɫ] mismatches. It includes predictors that capture both phonetic and social influences, specifically *milliseconds per syllable*, *PLUS (percent life in the US)*, *stress*, and *syllable type*.

In this model, *PLUS* ($\beta = .05$, $p = .004$) and *syllable type: closed* ($\beta = .83$, $p = .005$)

are particularly influential, both showing positive effects on mismatch likelihood. This indicates that a higher percentage of life spent in the US and closed syllables are associated with a greater chance of dark-l mismatch. Though *stress: unstressed* and *milliseconds per syllable* approached significance, their impacts are more modest ($p = .089$ and $p = .092$, respectively).

Table 5.6: Mixed Effects Model: Non-Velarized Lateral [l] Match vs. Velar Lateral [ɫ] Mismatch Prediction

	Dep. Variable: Mismatch				
Predictors	β	SE	CI 95 %	z-stat	p-value
Intercept	-9.42	1.48	[-12.32−-6.53]	-6.38	.000***
Milliseconds per Syllable	.18	.10	[-.03−.38]	1.68	.092*
PLUS	.05	.02	[.02−.09]	2.90	.004***
Stress *(ref:stressed)*					
Unstressed	.48	.29	[-.07−1.04]	1.70	.089*
Syllable Type *(ref:open)*					
Closed	.83	.29	[.26−1.41]	2.83	.005***
Random Effects					
σ^2	3.29				
τ_{00} Speakers	4.95				
ICC	.60				
N Speakers	22				
SD Speakers	2.22				
Observations	2509				
Marginal R^2/ Conditional R^2	.33 / .73				
AIC	504.71				

$^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$

Results shown in Figure 5·11 highlight *PLUS* and *syllable type* as having the largest impacts on model fit, with AIC increases of 6.4 for each if dropped. Meanwhile, *stress* and *milliseconds per syllable* had comparatively small effects, indicating their roles may be more peripheral in this specific context.

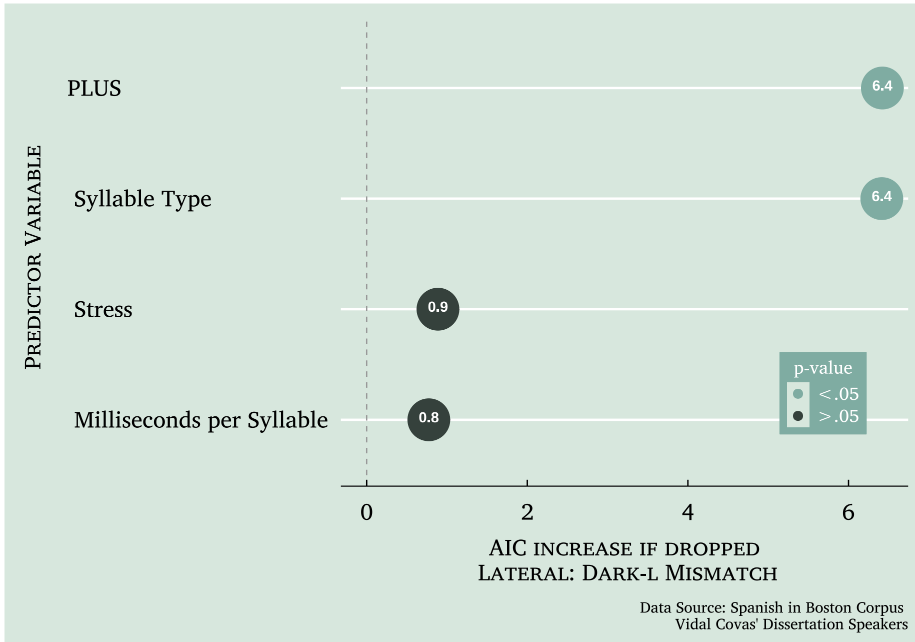

Figure 5·11: Impact of predictor variable removal on AIC — /l/: [l] & [ɫ]

The random effects show substantial variability between speakers, with an *ICC* of $.60$, suggesting that speaker-level differences account for 60% of the variance in dark-l mismatches. Additionally, the model's *marginal* r^2 is $.33$, while the *conditional* r^2 is $.73$, indicating that fixed effects alone explain 33% of the variance, with the full model explaining 73% when random effects are included.

5.2.5.3 Summary of the Results for Lateral Subset: Dark-L Mismatch

Among the factors influencing the likelihood of a lateral being realized as [ɫ] instead of [l], *PLUS* and *syllable type* emerge as the strongest predictors. Speakers who have spent a greater percentage of their lives in the U.S. are more likely to produce [ɫ] mismatches. Additionally, closed syllables favor the production of this mismatch, highlighting the role of syllable structure in the realization of laterals.

Although *stress* and *milliseconds per syllable* did not reach conventional significance thresholds, they still contribute to the model's overall fit. The direction of their effects suggests that unstressed syllables and shorter syllable durations (i.e., faster speech rates) may be associated with a higher likelihood of dark-l realization, though their influence is weaker compared to *PLUS* and *syllable type*.

The random effects indicate substantial speaker-level variation, with an *ICC* of .60, meaning that a significant portion of the variation in dark-l mismatch patterns is attributable to individual differences among speakers. This suggests that while linguistic and social predictors play a role, speaker-specific factors—such as idiosyncratic variation in articulation or sociolinguistic identity—may also contribute to the realization of this variant.

5.2.6 Summary of Findings Across Liquid Models

In this section, I summarize the key findings across the six models analyzed, focusing on the trill, tap, and lateral liquids, along with their respective subsets. Each model reveals unique insights into the predictors influencing mismatch rates and highlights both shared and distinct factors across liquid types. Table 5.7 presents an overview of the main effects across the models. Here, a " + " indicates an increased likelihood of mismatch, while a "-" corresponds to a decreased likelihood of mismatch. Cells shaded in green reflect predictors that showed significant effects. A pattern observed in the rhotic models is that higher lexical frequency tends to increase the likelihood of mismatch, aligning with prior research on frequency-driven phonetic reduction. This reflects broader usage-based findings that frequent forms are more likely to undergo reduction—such as intervocalic /d/ deletion in

Spanish (Bybee 2002) or phonetic shortening in Dutch affixes (Pluymaekers et al. 2005). However, this effect did not appear in the lateral models, underscoring that frequency alone is not a universal predictor of variation. This mirrors inconsistencies found in morphosyntactic variation research, where frequency sometimes acts as an amplifier of other constraints rather than exerting a direct effect (Poplack 2001; Erker & Guy 2012; Dionne 2023).

Beyond these frequency-related effects observed in the rhotic models, a range of articulatory, contextual, and prosodic factors—such as *syllable structure*, *position in the word or syllable*, *stress*, and *speech rate*—also shaped liquid variation to varying degrees across models. These findings align with prior research on segmental weakening and phonetic reduction in Spanish (Poplack 1979; Alba 1990; Quilis 1993), particularly in highlighting the greater susceptibility of coda liquids to mismatch. While *PLUS* did not significantly predict tap variation, it consistently emerged as a significant factor in the models for trill mismatches and in the subset of laterals realized as [ɫ], suggesting that increased U.S. life experience influences variability in certain liquid types. However, this effect does not indicate wholesale convergence with English patterns; rather, it suggests selective adaptation, with different liquids exhibiting distinct trajectories of stability and change.

For taps, the strongest social predictors were *country of origin* and *sex* rather than contact measures, pointing to the maintenance of dialectal patterns rather than direct influence from English. The clearest example of this is lateralization of the tap, where PLUS had no effect: a Puerto Rican male is significantly more likely to lateralize than not, underscoring the persistence of region-specific articulations rather than contact-driven change.

Table 5.7: Summary of Results Across Liquids Models

PREDICTOR VARIABLE	TAP	/ɾ/ - [l]	TRILL	/r/ - [r̝]	LATERAL	/l/ - [ɫ]
Log Lexical Frequency	+	+	+	+		
Milliseconds per Syllable	-	-	-	-		+
Position in Word *(ref:initial)*						
Internal			-	-		
Position in Syllable *(ref:onset)*						
Coda	+	+				
Previous Sound *(ref:cons)*						
Pause					-	
Vowel		+			-	
Following Sound *(ref:cons)*						
Pause	-	-			-	
Vowel	-	-			-	
Stress *(ref:stressed)*						
Unstressed	-	-			+	+
Syllable Type *(ref:open)*						
Closed					+	+
Country of Origin *(ref:DR)*						
Puerto Rico	+	+	-	-		
PLUS	+		+	+		+
% English Only					+	
Sex *(ref:female)*						
Male	+	+		-		

\+ indicates an increased likelihood of mismatch
— indicates a decreased likelihood of mismatch
shaded cells reflect predictors that showed significant effects: $p < 0.1$; $p < 0.05$; $p < 0.01$

In contrast, the trill and lateral models tell two distinct stories regarding contact effects. For the trill, *PLUS* is the only significant social predictor when considering the entire dataset. However, when narrowing the focus to the subset of trill tokens realized as [r̝], a more nuanced picture emerges. Here, *country of origin* and *sex* become moderately significant, revealing that a Dominican female who has spent a greater portion of her life in the U.S. is more likely to produce [r̝]—a variant that already exists in Spanish—than the canonical [r]. Rather than signaling English influence, this suggests that speakers are making a sociolinguistic choice to use an available Spanish variant, possibly as a means of indexing group belonging within a Caribbean Spanish identity.

The lateral, on the other hand, presents a clearer case of contact-induced variation, as the only significant social predictors are related to English exposure. Contact appears to drive the likelihood of lateral mismatches, with the subset analysis of dark-l realization further reinforcing this. Not only is *PLUS* a significant predictor in this subset, but it also has the strongest effect on model fit, highlighting the role of U.S. exposure in shaping lateral variation. Crucially, [ɫ] is not a variant that exists in Spanish outside of contact with English, distinguishing it from the other liquid variations examined. However, this does not mean that speakers are unaware of /l/ as a site of variation. Rather, it may be that /l/ does not carry the same weight in signaling national or regional belonging, leading speakers to devote less effort to maintaining a distinctively "Puerto Rican" or "Dominican" realization of the lateral while focusing more on the rhotics.

Taken together, these results suggest that while liquids are often discussed collectively as a variable feature of Caribbean Spanish, speakers appear to differentiate them in their sociolinguistic behavior. All three liquids exhibit variation, but rhotics seem to carry greater social significance, making them stronger candidates for identity marking within Caribbean Spanish. By contrast, /l/, which lacks the same social weight, may be more susceptible to bilingual optimization strategies, as speakers prioritize variation in features that provide greater social leverage. In other words, speakers may direct their effort toward linguistic variables that yield the most symbolic capital within their social and linguistic networks, while allowing /l/ to shift more freely under the influence of English contact.

6 Covariation and Coherence: Relationships between Linguistic Variables

As discussed previously, the study of linguistic variation in contact settings offers a unique lens through which to understand the interplay between salience, agency, and social meaning. Previous studies have emphasized the importance of examining relationships between variables to uncover patterns of structural convergence, resistance, or independence in bilingual communities (Erker & Otheguy 2016; Guy et al. 2022). Building on these insights, this chapter investigates the co-variation of the linguistic features in this study among Spanish-speaking Bostonians of Puerto Rican and Dominican origin, focusing on how salience potentially shapes linguistic behavior in a contact setting.

To assess covariation, I explored patterns of variation in five variables: *liquid variation* (n = 7,354), *coda /s/* (n = 3,806), *filled pauses* (n = 1,718), *subject personal pronouns* (n = 7,240), and *subject placement* (n = 4,079). By framing these variables within a continuum of salience, this analysis explores how linguistic features potentially differ in their relationship to the pressures of contact. The following sections detail the relationships among these variables and how they interact within the broader linguistic landscape under investigation.

This co-variation analysis directly addresses the second research question of the study: whether salient and non-salient features pattern differently in contexts of language and dialect contact. Variables on the lower end of the salience continuum, such as filled pauses and subject placement, show evidence of structural convergence with English, reflecting bilingual optimization strategies and the pressures of cognitive efficiency. In contrast, higher-salience variables like coda /s/ and liquids appear comparatively more stable across contact-related metrics. This relative stability is not due to any inherent resistance to change, but rather to the fact that salient features are more likely to be noticed, talked about, and strategically deployed by speakers. As such, they are more responsive to social rather than cognitive pressures—and their trajectories depend on the specific ideologies and identity goals of a given community. In the case of Spanish-speaking Boston, speakers exhibit strong own-group loyalty and high dialectal awareness, conditions that favor the maintenance of nationally recognizable speech patterns. Thus, the comparative patterning of convergence and stability observed here reflects a broader dynamic: while lower-salience features tend to shift in predictable, cognitively motivated ways, the treatment of higher-salience features is more variable, shaped by social meaning, ideological orientation, and local histories of contact.

As the final results chapter, this analysis, when considered alongside the findings from earlier chapters, provides an overview of how salience informs our understanding of linguistic variation and covariation in bilingual contact settings. By examining how features co-vary—or fail to co-vary—this chapter illuminates the mechanisms through which linguistic systems vary and change in contact settings. These insights are critical for understanding how speakers utilize their linguistic repertoires to construct identity, manage group affiliations, and respond to sociolinguistic pressures. The patterns observed in this chapter lay the groundwork for the subsequent discussion and conclusion chapter, where their broader implications for sociolinguistic theory, language contact, and the role of salience in shaping linguistic variation and change will be explored. Before examining the relationships between these features, Section 6.1 establishes their individual distributions.

6.1 Overview of Empirical Patterns

While the previous chapter provided an analysis of liquid variation, this section shifts focus to four additional variables that have not yet been discussed in detail: *coda /s/, filled pauses, subject personal pronouns, and subject placement*. Building on prior work in these four variables, this section presents a general overview of the distribution of their variants in the speech of the 22 Spanish-speaking Bostonians analyzed in this dissertation.

Subject Pronoun Expression The analysis of subject personal pronouns followed the methodology of Otheguy & Zentella (2012), categorizing eligible finite verb tokens as either *present* or *absent* with respect to pronominal forms. A pronoun was classified as *present* if it explicitly appeared in speech, while it was considered *absent* if it did not, as shown in Table 6.1. The dataset included a total of 7,240 sites of pronominal variation, of which 2,698 (37.27%) were categorized as *present* and 4,542 (62.73%) as *absent*.

Table 6.1: Examples of Subject Pronoun Expression Coding

(1) **Yo** bailo salsa en la plaza los jueves. También (2) **ø** bailo merengue. Los miércoles (3) **ø** tomo clases de gimnasia. Ignacio nada. (4) **Él** también alza pesas pero (5) **ø** no hace gimnasia.
(1) I dance salsa in the plaza on Thursdays. (2) I (ø) also dance merengue. On Wednesdays, (3) I (ø) take gym classes. Ignacio swims. (4) He also lifts weights but (5)He ødoesn't do gymastics.

Token #	Pronoun	Present/ Absent	Translation
(1)	yo	present	I
(2)	ø	absent	(I)
(3)	ø	absent	(I)
(4)	él	present	he
(5)	ø	absent	(he)

Subject Placement Subject placement was analyzed across both pronominal and non-pronominal subjects, following the approach used in previous studies (Erker et al. 2017; Erker 2017). Subjects were categorized based on word order: *pre-verbal* subjects followed *SV* order, while *post-verbal* subjects followed *VS* order, as illustrated in Table 6.2. Of the 4,079 subject tokens analyzed, 3,613 (88.58%) appeared in the *pre-verbal* position, while 466 (11.42%) were *post-verbal*.

Table 6.2: Examples of Subject Placement Coding

(1) **Ignacio** es muy rápido en natación, pero sus mejores estilos son la mariposa y el estilo libre, donde siempre gana medallas (2) **él**. En el relevo de estilos, (3) **Ignacio** nada la parte de estilo libre, mientras hacen pecho, mariposa y espalda (4) **sus compañeros**.
(1) Ignacio is very fast at swimming, but his best strokes are butterfly and freestyle, where (2) he always wins medals. In the medley relay, (3) Ignacio swims the freestyle part, while (4) his teammates do breaststroke, butterfly, and backstroke.

Token #	Subject	Pre/Post -verbal	Translation
(1)	Ignacio	pre-verbal	Ignacio
(2)	él	post-verbal	he
(3)	Ignacio	pre-verbal	Ignacio
(4)	sus compañeros	post-verbal	his teammates

Filled Pauses The methodology used for describing variation in filled pauses can be found in Erker & Bruso (2017) and Erker & Vidal-Covas (2022). As demonstrated in Table 6.3, tokens were categorized based on vowel quality, distinguishing between centralized and not centralized variants. Filled pauses containing [a] or [ə] were classified as *centralized*, while those containing [e] were classified as *not centralized*. Across the 22 interviews, a total of 1,718 filled pauses were identified, with 999 (58.15%) categorized as *centralized* and 719 (41.85%) as *not centralized*.

Table 6.3: Examples of Filled Pause Categorization

(1) **[e]** Yo entreno todos los días porque quiero mejorar mi tiempo y (2) **[e]** ganar más medallas. (3) **[əm]** A veces estoy muy cansado, (4) **[ə]**, pero (5) **[ə]** como quiera (6) **[am]** voy a la piscina y (7) **[a]** entreno.
(1) Eh I train every day because I want to improve my time and (2) eh win more medals. (3) Um sometimes I'm really tired, (4) uh, but (5) uh anyway (6) um I go to the pool and (7) uh train.

TOKEN #	VOWEL	CENTRALIZATION	TRANSLATION
(1)	[e]	not centralized	eh
(2)	[e]	not centralized	eh
(3)	[əm]	centralized	um
(4)	[ə]	centralized	uh
(5)	[ə]	centralized	uh
(6)	[am]	centralized	um
(7)	[a]	centralized	uh

Coda /s/ The classification of coda /s/ in this dissertation built on the methodological foundation established in Erker (2017) and Erker & Reffel (2021), but introduced a novel coding approach. Specifically, a *match/mismatch* framework—like the one used in the analysis of liquid variation—was employed to evaluate whether the presumed PHONOLOGICAL FORM and observed SURFACE FORM matched. If the surface realization was anything other than [s], it was classified as a *mismatch*. This includes realizations such as [z], [h], [Ø] (deletion), or less common variants like [g]. In contrast, tokens that retained [s] as their surface form were classified as *match*. Table 6.4 provides examples of this classification. Across the dataset, a total of 3,806 tokens of coda /s/ were analyzed, with 2,071 (54.41%) classified as *match* and 1,735 (45.59%) as *mismatch*.

Table 6.4: Examples of Match/Mismatch Dependent Variable for Coda /s/

HOST WORD	TRANSLATION	PHONOLOGICAL FORM	SURFACE FORM	MATCH/ MISMATCH
de***s***pertar	to wake up	/s/	[s]	MATCH
de***s***proporcionado	disproportionate	/s/	[∅]	MISMATCH
e***s***cuela	school	/s/	[g]	MISMATCH
e***s***tilo	style	/s/	[h]	MISMATCH
ine***s***tabilidad	instability	/s/	[s]	MATCH
re***s***ponsabilidad	responsibility	/s/	[z]	MISMATCH
tran***s***cripción	transcription	/s/	[s]	MATCH

With these baseline distributions established, we now turn to the relationships between these variables.

6.2 Variation in Lower Salience Variables in a Contact-Setting

Keeping in mind the discussion of Spanish-English bilingualism and language use patterns presented in Section 4.2, let us explore how low-salience features areused in the contact setting of interest. Specifically, I reconstruct previously published analyses of three variants —pronoun presence, pre-verbal subjects, and centralized filled pauses—to provide context within the broader sample of 80 speakers from the Spanish in Boston Corpus. These variables have been shown in prior studies to exhibit patterns of change that, while modest, increasingly align their usage with the grammatical properties of analogous structures in English (Erker 2012). The broader patterns observed among these speakers provide a foundation for interpreting the behavior of the 22 speakers analyzed in detail throughout this dissertation.

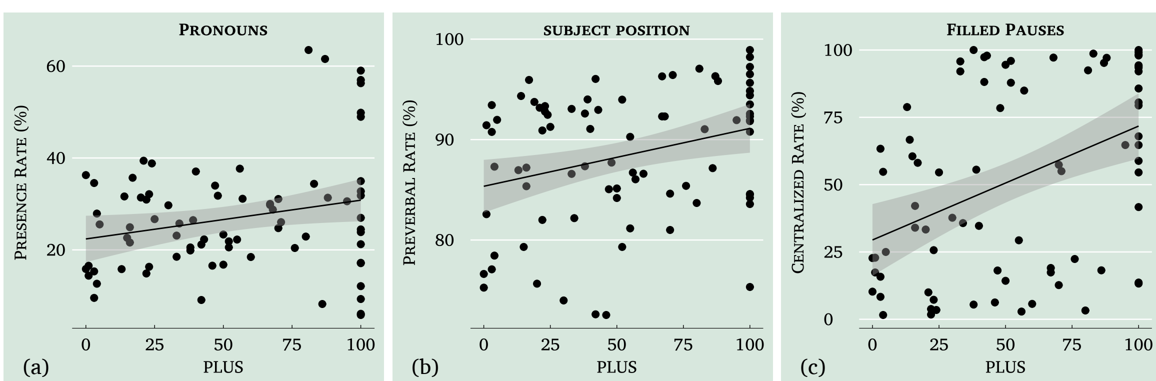

Figure 6·1: Relationship of pronoun presence, pre-verbal subjects, and centralized filled pauses to PLUS among 80 Bostonians. These three figures represent a larger sample from the Spanish in Boston Corpus, compared to the 22 speakers analyzed in most parts of the dissertation.

Figure 6·1 highlights the trends in the use of these features among the 80 speakers.

The specific patterns observed are as follows:

PRONOUN PRESENCE (6·1A) - Higher rates of pronoun presence are associated with an increase in *Percent Life in the United States (PLUS)*, reflecting structural alignment with English, where pronoun use is nearly obligatory.

PRE-VERBAL SUBJECTS (6·1B) - Speakers with greater mainland U.S. residency demonstrate a higher frequency of pre-verbal subjects, a characteristic feature of English syntax.

CENTRALIZED FILLED PAUSES (6·1C) - The use of centralized filled pauses increases alongside PLUS, further supporting the evidence of cross-linguistic influence.

6.2.1 Zooming In: Low-Salience Features Among 22 Speakers

To more closely examine the linguistic behavior of the 22 Spanish-speaking Bostonians analyzed in this dissertation, Figure 6·2 provides a zoomed-in view of three lower-salience features—pronoun presence, subject position, and centralized filled pauses—in relation to Percent Life in the United States (PLUS). The patterns observed in this subset generally mirror the broader trends seen in the larger corpus: as speakers' time in the U.S. increases, their usage of these features shifts in ways that more closely align with English norms.

Specifically, speakers with greater U.S. residency tend to use pronouns more frequently (Figure 6·2a), favor pre-verbal subject constructions (Figure 6·2b), and show increased use of centralized filled pauses (Figure 6·2c)—features that parallel English usage. In the case of filled pauses, speakers increasingly favor the centralized variant [a], which is available in both English and Spanish, potentially as a way to reduce processing costs in bilingual

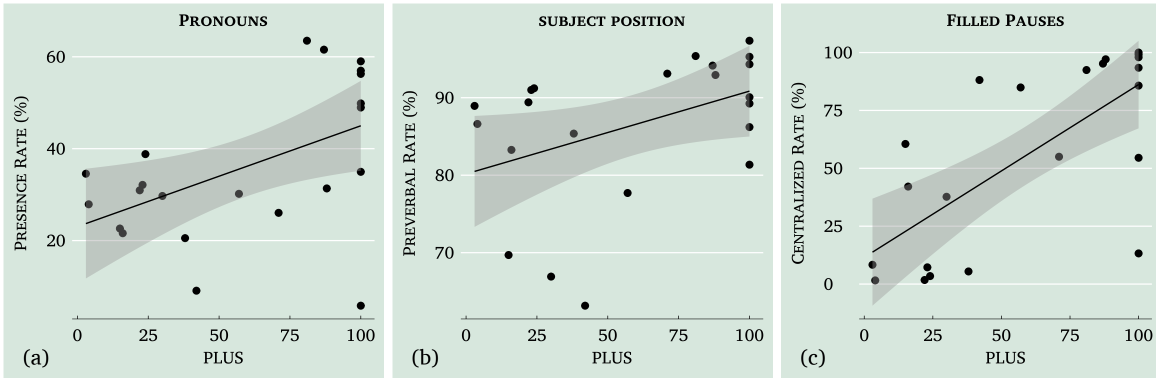

Figure 6·2: Relationship of pronoun presence, pre-verbal subjects, and centralized filled pauses to PLUS among 22 Spanish-speaking Bostonians. These three figures focus on the 22 speakers analyzed in detail throughout the dissertation.

speech. These observations are consistent with prior findings that socially weak variables may shift in ways that ease cross-linguistic processing, potentially as part of bilingual optimization strategies (Muysken 2013; Erker & Bruso 2017).

These patterns, while suggestive, should be interpreted cautiously. Given the absence of a full constraints-based analysis, we cannot determine whether these surface-level trends reflect changes in underlying grammars or shifts in usage conditioned by other social or linguistic factors. To probe this further, the next section presents regression analyses at the token level, offering a more fine-grained account of how social predictors like U.S. residency correlate with the use of low-salience variables.

Regression Model Results: Low-Salience Features Among 22 Speakers

The patterns observed are supported by regression model results, summarized in Table 6.5. These analyses are conducted at the token level, examining individual instances of pronoun presence/absence, subject positioning, and pause filling rather than aggregate speaker-level patterns. The table presents findings for *pronoun presence*, *pre-verbal subjects*, and *centralized filled pauses* among the 22 Spanish-speaking Bostonians analyzed.

Unlike the liquids chapter, these models focus exclusively on social predictors, rather than linguistic ones. This reflects the fact that linguistic constraints vary across variables, making direct comparisons difficult. More importantly, the aim of this analysis is to examine how social factors shape patterns of linguistic variation and covariation in a contact setting, rather than to model internal linguistic constraints. By centering social influences, this approach highlights the ways in which external factors structure speakers' linguistic behavior across multiple variables.

Table 6.5: Summary of Model Results Corresponding to Pronoun Presence, Pre-Verbal Subjects, and Centralized Filled Pauses among 22 Spanish-speaking Bostonians

Predictors	Stat	Dependent Variable: Pronouns Present[1]	Preverbal Subjects[2]	Centralized FPs[3]
Intercept	β	$-.62$	1.58^{**}	-1.43
	SE	(.59)	(.67)	(1.77)
Country of Origin *(ref:DR)*				
Puerto Rico	β	$-.12$	.03	.06
	SE	(.29)	(.31)	(.89)
PLUS	β	$.01^{**}$	$.02^{***}$	$.05^{***}$
	SE	(.00)	(.00)	(.01)
Age	β	$-.01$	.00	$-.03$
	SE	(.01)	(.01)	(.04)
Sex *(ref:female)*				
Male	β	$-.49^{*}$	$-.12$	$-.57$
	SE	(.29)	(.30)	(.86)
Random Effects				
SD $_{\text{Speakers}}$		.64	.59	1.87
ICC		.29	.26	.78
Observations		7240	3888	1718
Marginal R^2/ Conditional R^2		.06/.17	.09/.18	.44/.73

$^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$

Model Specifications:

[1] `glmer(PRONOUNS PRESENT ~ COUNTRY OF ORIGIN + PLUS + AGE + SEX + (1|SPEAKER))`

[2] `glmer(PREVERBAL SUBJECTS ~ COUNTRY OF ORIGIN + PLUS + AGE + SEX + (1|SPEAKER))`

[3] `glmer(CENTRALIZED FPs ~ COUNTRY OF ORIGIN + PLUS + AGE + SEX + (1|SPEAKER))`

Key results include:

PLUS - A significant positive effect on all variables, indicating that increased U.S. residency aligns with greater structural convergence to English.

COUNTRY OF ORIGIN - No significant effect, with variation largely driven by PLUS rather than national origin.

RANDOM EFFECTS - Notable variation across individual speakers, reflected in the random effects and intraclass correlation coefficients (ICC).

MODEL FIT - The models vary in explanatory power:

PRONOUN PRESENCE - *marginal* $r^2 = .06$, *conditional* $r^2 = .17$

PREVERBAL SUBJECTS - *marginal* $r^2 = .09$, *conditional* $r^2 = .18$

CENTRALIZED FILLED PAUSES - *marginal* $r^2 = .44$, *conditional* $r^2 = .73$

These values indicate that while speaker-level variation (captured by *conditional* r^2) accounts for a substantial portion of the variance, fixed effects alone (*marginal* r^2) explain a smaller but meaningful proportion, particularly for centralized filled pauses.

These results provide further evidence that, among the speakers analyzed, features at the lower end of the salience continuum exhibit parallel patterns of change in the same direction—namely, structural convergence with English—highlighting a shared trajectory of adaptation in this contact setting.

6.3 Salient Features as Tools for Speaker Identity

Speakers' treatment of coda /s/ is analyzed across the full sample of 80 speakers and the subset of 22 speakers who are of primary interest in the present study. Figure 6·3a highlights that while there is variation in coda /s/ realization across both the overall and subset groups, its treatment is more consistent and less related to PLUS compared to low-salience features. Among the 22 speakers (Figure 6·3b), patterns in coda /s/ realization remain tightly linked to national dialect norms, underscoring its salience and availability for speakers to use in their construction of identity. The analysis of coda /s/ mismatch

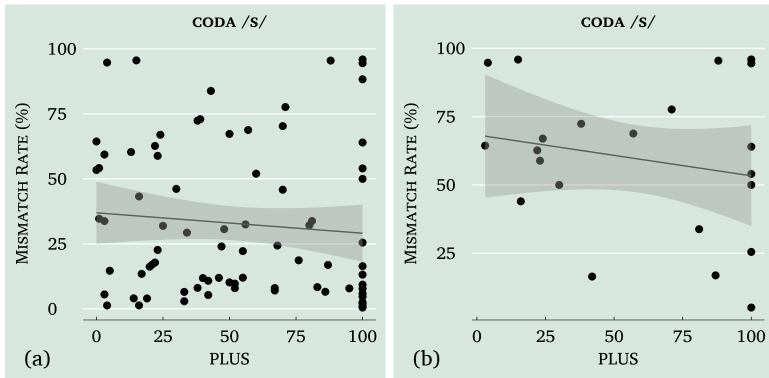

Figure 6·3: Comparison of the relationship between coda /s/ mismatch and PLUS in 80 and 22 Spanish-speaking Bostonians

among both the 80 and 22 speakers highlights the influence of national origin and the stability of this feature against contact-induced change. However, while coda /s/ provides valuable insights into dialectal maintenance and resistance to English structural convergence, it represents only one dimension of the broader picture of linguistic mismatch in these speakers. To more fully understand how these speakers use salient features in this context of language contact, it is crucial to consider all three features—coda /s/, tap-lateral mismatch, and trill-fricative trill mismatch—together.

Unlike coda /s/, the liquids were only analyzed in detail among the 22 speakers, representing one of the main contributions of this study. This focus enables a more comprehensive examination of how these features are associated with predictors like *PLUS* and *Country of Origin*, shedding light on the unique patterns of variation and stability that emerge within the group. The results for all three liquids are presented in Figure 6·4 and Table 6.6.

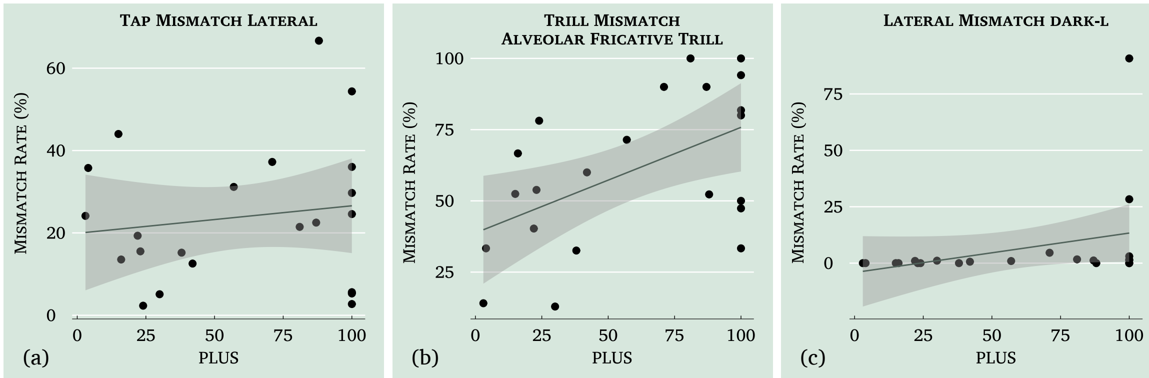

Figure 6·4: Relationship of tap mismatch, trill mismatch, and lateral mismatch to PLUS among 22 Spanish-speaking Bostonians

Table 6.6: Summary of Model Results Corresponding to Coda /s/ Mismatch, Tap-Lateral Mismatch, Trill-Fricative Trill, and Lateral Velarization Mismatch among 22 Spanish-speaking Bostonians

		Dependent Variable: Mismatch			
Predictors	**Stat**	**Coda /s/**	**/ɾ/ - [l]**	**/r/ - [r̝]**	**/l/-[ɫ]**
Intercept	β	$-.89$	-3.27^{***}	$.07$	-9.57^{***}
	SE	(1.30)	(.75)	(1.01)	(2.68)
Country of Origin *(ref:DR)*					
Puerto Rico	β	1.64^{**}	1.43^{***}	$-.61$	$-.99$
	SE	(.64)	(.38)	(.50)	(1.17)
PLUS	β	$-.01$	$.00$	$.02^{***}$	$.06^{***}$
	SE	(.01)	(.01)	(.01)	(.02)
Age	β	$.03$	$.02$	$-.00$	$.02$
	SE	(.03)	(.02)	(.02)	(.05)
Sex *(ref:female)*					
Male	β	$.03$	$.43$	$-.69$	$.82$
	SE	(.63)	(.37)	(.48)	(1.18)
Random Effects					
SD $_{\text{Speakers}}$		1.41	.79	.94	2.08
ICC		.67	.38	.47	.81
Observations		3806	3244	735	2509
Marginal R^2/ Conditional R^2		.13/.46	.12/.26	.17/.34	.37/.73

$^{*}p < .1$; $^{**}p < .05$; $^{***}p < .01$
Model Specifications:
`glmer(MISMATCH ~ COUNTRY OF ORIGIN + PLUS + AGE + SEX + (1|SPEAKER))`

The results presented in Table 6.6 shed light on the relationship between linguistic mismatch patterns and key predictors among 22 Spanish-speaking Bostonians. The findings suggest distinct influences for each dependent variable:

CODA /S/ MISMATCH (6·3B) - *Country of origin* is a significant predictor ($\beta = 1.64, p < 0.1$), indicating that speakers from Puerto Rico are more likely to exhibit mismatch. This suggests a maintenance of linguistic patterns tied to nation-state varieties rather than convergence with English structures. *PLUS*, however, does not significantly predict mismatch in this feature ($p > 0.1$), suggesting that coda /s/ does not exhibit the same pattern of change observed in lower-salience features. Model fit statistics show a moderate contribution of fixed effects (*marginal r*^{2} = .13), while speaker-level variation accounts for a substantial portion of the variance (*conditional r*^{2} = .46), highlighting the role of individual differences in this feature.

TAP-LATERAL MISMATCH (6·4A) - *Country of origin* also plays a significant role ($\beta = 1.43, p < 0.01$), with speakers from Puerto Rico showing greater mismatch. This further supports the idea of dialectal maintenance linked to national origin. *PLUS* does not emerge as a significant predictor ($p > 0.1$) for this feature, suggesting that contact with other dialects of Spanish—whether through interaction with non-Caribbean speakers or exposure to standard norms—has not led to reconfiguration of this feature. The model explains a moderate amount of variance through fixed effects (*marginal r*^{2} = .12), with speaker-level variation contributing substantially (*conditional r*^{2} = .26), suggesting that individual speaker tendencies shape patterns of variation in this feature.

TRILL-FRICATIVE TRILL MISMATCH (6·4B) - *PLUS* has a significant positive effect ($\beta = 0.02, p < 0.01$). While this might initially be interpreted as a sign of structural alignment with English or possibly dialectal leveling, a closer analysis of the specific variant used by speakers reveals an alternative explanation. Speakers with higher PLUS values are more likely to use a trill variant that is not found in English but is attested within the broader Spanish linguistic repertoire. Rather than signaling convergence, this pattern may represent a nuanced form of linguistic resistance—one in which adaptation to mainland U.S. residency is accompanied by the reinforcement of variation internal to Spanish. In this way, speakers may maintain aspects of their linguistic identity even as they navigate life in a bilingual environment. The model captures a portion of the variance through fixed effects (*marginal* $r^2 = .17$), while speaker-level variation remains influential (*conditional* $r^2 = .34$), indicating that both social factors and individual differences contribute to this pattern.

LATERAL-VELARIZED LATERAL MISMATCH (6·4B) - *PLUS* has a significant positive effect ($\beta = 0.06, p < 0.01$). As discussed in the chapter dedicated to the analysis of liquids, the [ɫ] variant is absent in monolingual Spanish contexts and emerges primarily in the context of contact with English. This sets it apart from the subsets of the other two liquids studied. Nonetheless, speakers recognize /l/ as a variable feature. It appears that /l/ lacks significant social indexing of national or regional identity, leading speakers to invest less effort in preserving a distinctly 'Puerto Rican' or 'Dominican' pronunciation of the lateral, while placing greater emphasis on rhotic sounds. The model demonstrates a strong ability to account for variation in this feature, with fixed effects explaining a substantial portion of the variance (*marginal* $r^2 = .37$) and speaker-level variation playing an even greater role (*conditional* $r^2 = .73$), suggesting that individual differences strongly shape patterns of lateral variation.

6.4 Intermediate Conclusion: Salience and Covariation

The results presented in this chapter demonstrate that features at the higher end of the salience continuum do not exhibit the same patterns of covariation observed among lower-salience variables, which show more consistent intergenerational trends. Among the low-salience features—*pronoun presence*, *pre-verbal subjects*, and *centralized filled pauses*—there is clear evidence of parallel change: each shows a significant positive relationship with *PLUS*, suggesting that these features are undergoing rate-level change in comparable directions. These patterns reflect systematic shifts in usage that align with structures found in English, likely facilitated by their low social visibility and reduced indexical potential.

In contrast, *coda /s/* shows no evidence of change with respect to PLUS, and remains closely tied to national origin. Among the liquid variables, the results are more mixed: while *lateral velarization*([ɫ]) shows signs of contact-induced restructuring, *trill-fricative trill* alternation appears to reflect a more complex form of adaptation—one that resists convergence with English and draws instead from the broader Spanish linguistic repertoire. Notably, *tap-lateral alternation*, the highest-profile of the liquid variables, shows no evidence of change induced by either language or dialect contact, and is best predicted by national origin.

Taken together, these findings suggest that features at the higher end of the salience continuum do not exhibit consistent patterns of covariation in response to contact. Rather than shifting uniformly, these features appear to offer speakers greater flexibility in how they are used. Speakers seem to selectively navigate their linguistic choices, balancing the pressures of contact with a commitment to Spanish-specific variants. Some salient variables, such as *coda /s/* and *rhotics*, carry greater social weight and are more likely to be actively maintained or strategically deployed to signal identity. In contrast, /l/, which may lack the same indexical significance, appears more susceptible to contact-driven restructuring—perhaps because speakers invest less effort in maintaining national or dialectal norms for this feature. These contrasts underscore the importance of salience not only in shaping patterns of change, but also in enabling speakers to navigate and negotiate linguistic variation in bilingual settings.

7 Discussion & Conclusions: Linguistic Salience and Contact-Induced Change

Linguistic variation in contact settings presents an especially interesting opportunity to examine the interplay between salience, agency, and social meaning. The present study investigates this dynamic by analyzing patterns of variation in and potential relationships between multiple linguistic features in Spanish-speaking Bostonians. Two overarching research questions guided this study:

1. How do linguistic and social factors shape variation in the production of liquids in Spanish?
2. Do the data support the hypothesis that features at the lower end of the salience continuum and those at the higher end pattern differently in their relation to various metrics of contact experience?

The findings provide strong evidence that lower-salience features (e.g., *subject pronouns, subject placement, filled pauses*) exhibit structural convergence with English as an outcome of bilingual optimization strategies aimed at maximizing common ground across linguistic systems. Conversely, higher-salience features (e.g., *coda /s/* and *liquid variation*) tend to show greater stability—not because they are inherently resistant to change, but because they are governed by different processes in contact settings. Rather than reflecting cognitively motivated—and potentially subconscious, or at least non-socially motivated—alignment across linguistic systems, these features function as socially charged resources. Speakers can—but do not always—strategically manipulate them to assert in-group membership, maintain dialectal identity, or resist external linguistic pressures. While this distinction between high- and low-salience features is well-documented in the literature (Erker 2017; Erker 2022; Guy et al. 2022), this study expands on previous work by incorporating liquid variation as a key sociolinguistic variable.

In what follows, I first summarize the findings on liquid variation and situate them within existing research. I then discuss patterns of covariation and coherence, focusing on the structural convergence of features on the lower end of the salience spectrum and the social significance of higher-salience features. Finally, I reflect on the broader implications of these findings, emphasizing how certain salient features afford greater indexical potential than others, and how speakers navigate this variation in bilingual contact settings.

7.1 Linguistic and Social Conditioning of Liquid Variation

The regression analyses presented in Chapter 5 reveal that the three Spanish liquids—taps, trills, and laterals—are not uniformly conditioned by linguistic and social factors. Instead, each liquid displays a distinct profile, shaped by a combination of articulatory, phonological, lexical, and sociolinguistic influences. These patterns not only echo previous findings in the literature but also add nuance by disaggregating effects across subsets of the data and exploring how these predictors interact in bilingual, urban contexts.

7.1.1 Taps: Strong Internal Conditioning, Dialect-Based Social Patterning

Tap variation is strongly constrained by internal linguistic factors. The most robust predictors of mismatch between presumed phonological form and observed phonetic realizations included syllable position, phonological context, lexical frequency, and speech rate. These findings align with prior studies showing that consonants in coda position are more susceptible to reduction or reanalysis (Poplack 1979; Alba 1990; Quilis 1993; Ramos-Pellicia 2004; Beaton 2016). The influence of following segment also mirrors classic accounts of coarticulatory influence on liquid realization, with taps more stable when flanked by vowels.

Notably, the data also reveal that taps are more likely to be realized canonically in prosodically weaker (i.e., unstressed) syllables—contrary to typological expectations, where reduction typically favors weaker positions. While—so far—limited to the present data, this finding suggests a potential divergence in Spanish tap behavior that warrants further investigation.

Importantly, social factors also play a role in shaping tap variation. Country of origin emerged as a significant predictor, with Puerto Rican speakers more likely than Dominicans to produce mismatches, especially lateralized taps—a hallmark feature of Puerto Rican Spanish. This is consistent with previous descriptions of lateralization in the Caribbean, particularly in Puerto Rico (Lipski 2012; Shenk 2021). Gender was also significant, with male speakers more prone to mismatch, echoing past findings that associate men with less standard or more locally marked speech forms (Beaton 2005).

Interestingly, contact-related measures, such as PLUS, did not significantly predict variation for taps, suggesting that tap lateralization persists primarily as a dialect-internal feature, resistant to contact-driven restructuring. These results reinforce the idea that taps are maintained as a regionally specific variant, particularly among Puerto Rican speakers, and are less affected by English influence or inter-dialectal interaction.

7.1.2 Trills: Bilingual Contact as a Primary Driver

Unlike taps, trill variation appears more sensitive to bilingual exposure. The most robust social predictor in the overall trill model was Percentage of Life in the U.S. (PLUS), which positively predicted mismatch. This suggests that speakers with more extensive U.S. residency are more likely to deviate from the canonical trill, often producing the variant [r̝], a fricative-like realization previously attested in Caribbean Spanish varieties (Delgado-Díaz & Galarza 2015).

However, a closer look at the mismatch forms complicates this interpretation. The most frequent non-canonical realizations are not borrowed from English, but rather emerge from within the broader inventory of Spanish phonetic variation—especially those common in Caribbean varieties. This suggests that the observed patterns may not reflect straightforward convergence with English phonology, but instead result from contact with other Spanish dialects. As such, the correlation between trill variation and U.S. residency likely indexes increased exposure not only to English, but also to Spanish varieties beyond a speaker's own dialect background—especially in an urban environment like Boston.

This interpretation is supported by the trill subset model, where both PLUS and lexical frequency showed significant effects, and speaker gender and national origin emerged as marginally significant. Taken together, these findings suggest that bilingual contact, in the broadest sense, shapes trill realization—not solely through structural influence from English, but through the cumulative effects of multilingual, multi-dialectal interaction. The fricative trill variant may thus be reanalyzed or reweighted in its salience or acceptability through this contact-driven exposure, further underscoring the complex dynamics of variation in diaspora communities.

7.1.3 Laterals: Contact-Induced Restructuring and Dialect Mixing

Among the three liquid phonemes, the lateral /l/ shows the clearest evidence of contact-induced restructuring. In contrast to both rhotic liquids, lateral variation (/l/) showed minimal alignment with national origin and was shaped primarily by contact with English and increased exposure to diverse Spanish dialects. In the lateral models, social predictors such as % English-only interaction with interlocutors and PLUS (Percentage of Life in the U.S.) emerged as significant, while national origin had no effect. This suggests that variation in /l/ is less tied to national identity and more reflective of bilingual and bidialectal interaction in U.S. urban environments.

The treatment of the lateral reflects broader linguistic experiences shaped by life in the United States—including not only increased contact with English, but also greater interaction among Spanish speakers from diverse dialect backgrounds. This interpretation aligns with prior work on L2 Spanish, such as Solon (2015), which shows that English-speaking learners often produce velarized [ɫ] in coda position due to transfer from English. Solon also notes that this cue significantly contributes to the perception of a foreign accent. While her study focuses on L2 learners, the current findings suggest that a similar phonetic influence may emerge in heritage and bilingual speakers as a result of regular use of

English.

While [ɫ], a variant not associated with Caribbean Spanish but common in English, was the most frequent mismatch form in the dataset (6.7%), the overwhelming majority of lateral tokens (88.9%) were produced as canonical [l]—a variant widely preferred across all Spanish varieties. The presence of velarized /l/ appears to correlate with increased mainland U.S. life experience, likely reflecting cross-linguistic influence. However, it remains a minority realization, suggesting that even in contexts of sustained bilingualism, canonical /l/ continues to dominate production patterns. Unlike some L2 contexts where [ɫ] becomes a dominant or default realization, the speakers in this study do not exhibit wholesale convergence with English norms. Instead, their productions suggest that even in contexts of intense bilingualism, speakers continue to produce /l/ in ways consistent with Spanish phonological patterns. This, in turn, supports the idea that /l/, while arguably less socially charged than rhotics, is shaped more by language-internal phonological structure than by the conscious stylistic or social choices of speakers.

Moreover, the broader distribution of /l/ variants points to the possibility that dialect contact is also shaping variation. While processes such as deletion, aspiration, and vocalization—widely attested in Caribbean Spanish (Lipski 2008; Quilis 1993; Alba 1990; Alba 2004)—are present in the dataset, they are relatively rare, accounting for just 4.5% of tokens. This imbalance suggests a reconfiguration of how /l/ is socially interpreted in diaspora contexts, where contact with English and non-Caribbean Spanish varieties may be introducing new phonetic alternatives. As such, /l/ mismatch in this dataset appears to stem not from the preservation of inherited dialect norms, but from contact-induced restructuring in an urban, multilingual environment.

7.2 Covariation in Contact Settings: Structural Convergence vs. Sociolinguistic Indexing

This study reveals a striking asymmetry in the behavior of sociolinguistic variables in a contact setting: lower-salience features tend to shift together in systematic ways—showing shared alignment with contact-related metrics like PLUS—while higher-salience features show more differentiated patterning, often uncorrelated with those same metrics. This aligns with and extends existing models of covariation, coherence, and salience in bilingual communities (Erker & Otheguy 2016; Guy & Hinskens 2016; Erker 2022). Lower-salience features exhibit strong covariation and structural convergence with English—albeit modest in effect size—consistent with bilingual optimization strategies that prioritize communicative efficiency and cross-linguistic alignment. Rates of subject pronoun expression and preverbal subject placement increase significantly with U.S. mainland residency, reflecting alignment with English syntactic norms. Similarly, filled pauses shift toward centralized variants (e.g., [ə] or [a]), further indicating a systematic push toward structures that are more congruent across the speakers' two languages. Higher-salience features, by contrast, show more differentiated patterning, shaped less by structural convergence and more by social meaning. Coda /s/ demonstrates robust dialectal maintenance, with no significant correlation to length of U.S. mainland residency, suggesting its continued role as an index of ethnonational identity. Liquid variation reveals a more complex picture. While rhotics function as sociolinguistic boundary markers closely tied to national origin, laterals appear more stable overall. The presence of velarized /l/—a form not typical of Caribbean Spanish but common in English—does suggest the possibility of contact-induced restructuring; however, this remains a minority pattern in the dataset. There is also tentative evidence that intra-Spanish dialect contact may be contributing to this variation, though further research is needed to confirm this interpretation.

These findings underscore a core theoretical insight: salience shapes the coherence of linguistic behavior. Building on Labov's (1972) taxonomy of indicators, markers, and stereotypes, and Trudgill's (1986) work on awareness in contact settings, this study provides further support for the idea that features at the lower end of the salience continuum behave more "coherently"—i.e., they pattern together under bilingual pressure. In contrast, higher-salience features show a different kind of coherence—rather than shifting in tandem with contact-related pressures, they remain stable across metrics like PLUS, reflecting strong language and dialectal loyalty. This shared pattern suggests that, in this dataset, speakers treat salient variables not as markers of idiosyncratic expression but as stable components of group-linked identity—decoupling them from the shaping pressures of cognitive realignment in favor of maintaining their social significance.

This pattern highlights the central role of social meaning in shaping contact outcomes. Lower-salience variables tend to shift in similar ways, guided by bilingual optimization strategies. In contrast, salient variables such as coda /s/ and rhotics exhibit patterned stability—not because they are inherently resistant to change, but because speakers orient to them as key stylistic resources. As developed further below, it is through these socially prominent features that speakers engage in both accommodation and differentiation, mobilizing them to align with or distinguish themselves from interlocutors depending on context, stance, and social goals.

This interpretation aligns with and extends previous work by Erker & Otheguy (2016), Erker (2022), and Erker & Vidal-Covas (2024), which argues that non-salient features in

Spanish are more susceptible to structural convergence in contact settings, while salient features are more likely to persist. The current study corroborates this framework while adding nuance through the analysis of liquid variation. By showing that rhotics and laterals play distinct sociolinguistic roles—with the former indexing national origin and the latter reflecting linguistic experience and possible dialect contact—this study demonstrates that not all salient variables are treated identically. Their indexical meanings and stylistic affordances vary across features, groups, and contexts. In short, salience is not merely a cognitive or perceptual property—it is socially constructed and dynamically deployed. It plays a crucial role in mediating whether linguistic forms converge or persist, revealing the layered and agentive nature of variation in multilingual communities like Spanish-speaking Boston.

7.2.1 The Social Power of Salient Variables: In-Group Membership and Indexicality

The fact that speakers maintain dialectally particular patterns of liquid variation and coda /s/ while shifting lower-salience features suggests that salient variables provide speakers with sociolinguistic agency. This section explores how that agency manifests—specifically, how speakers use these features to index identity, assert dialectal distinctiveness, and position themselves within contact-induced linguistic change. While the previous section highlighted broader patterns of alignment and divergence, the focus here shifts to the social meanings that drive those patterns. Speakers can—and sometimes do—maintain highly salient variables in ways that reflect in-group alignment and ideological stance. But just as importantly, the socially charged nature of these forms makes their treatment more variable and context-dependent. The data suggest that such features are mobilized to signal belonging and distinction—often in nuanced and ideologically charged ways.

Liquid variation offers a clear window into this process. Among the two liquids examined, rhotics serve as powerful boundary markers between Dominican and Puerto Rican speakers. Because rhotic mismatch patterns align more strongly with national origin than with U.S. residency, they function as symbolic resources—akin to linguistic shibboleths—that reinforce dialectal distinctiveness and signal in-group membership (Valentín-Márquez 2007; Shenk 2021; Mendoza-Denton 2008). Their persistence, even in contexts of intense language and dialect contact, reflects the depth of their embeddedness in identity work.

Laterals, however, follow a different trajectory. Although also perceptually salient and frequently referenced by speakers, /l/ variation does not appear to index national identity in the same way. Instead, it reflects more diffuse linguistic experiences, including prolonged exposure to English and contact with speakers from diverse Spanish dialect backgrounds. In this way, lateral variation seems to index broader bilingual and pan-dialectal orientations, rather than ethnonational affiliation.

The contrast between rhotics and laterals underscores a key point: salience alone does not determine indexical function. Not all salient features carry the same social weight—or are interpreted in the same way by all speakers. While both coda /s/ and liquids are perceptually salient and frequently identified in metalinguistic commentary, the data show they operate differently in terms of indexical function and speaker control. Chapter 4 established that these two variables were explicitly referenced by speakers. Participants offered detailed descriptions of rhotic and lateral variation across regional varieties, as well as widespread commentary on /s/-weakening. These features were often invoked spontaneously and with considerable phonetic detail, underscoring their elevated status in community language ideologies. These patterns of metalinguistic awareness reinforce the idea that these features are socially meaningful and salient, not just to researchers but to speakers themselves. In this respect, the patterns observed among speakers in this study echo foundational insights from early sociolinguistic work—particularly Labov's studies of Martha's Vineyard (Labov 1963) and New York City (Labov 1966; Labov 1972)—which demonstrated that linguistic variation can carry social meaning and align with group identity, even when speakers are not consciously aware of their own behavior. Crucially, those studies took place outside of language contact settings, underscoring that this kind of socially embedded variation is not unique to bilingual contexts. The present findings thus instantiate a dynamic that spans both contact and non-contact settings, while also showing how contact adds further layers of social meaning and complexity to the variation. In cases of heightened salience, such features may also become available for stylistic manipulation, particularly when they intersect with salient social categories.

These ideological stances are not simply abstract—they have statistical correlates in the production data. To further explore the relationship between social meaning and the use of salient features, I conducted follow-up regression analyses focusing on coda /s/ and tap lateralization—the two variables that showed strong effects for national origin in the covariation chapter. Building on that finding, and given the prominence of these features in both production patterns and metalinguistic commentary, I ran separate models for Puerto Rican and Dominican speakers to examine how their linguistic behavior related to their attitudes about national identification. Specifically, I analyzed how Puerto Rican speakers responded to being identified as Puerto Rican, and how Dominican speakers responded to being identified as Dominican—whether they liked it, disliked it, or felt indifferent—and whether these stances were associated with differing patterns of linguistic behavior. This approach allowed for a more nuanced understanding of how stance and identity positioning influence the maintenance or shifting of salient dialect features.

Among Puerto Rican speakers, those who reported liking being identified as Puerto Rican were significantly more likely to exhibit mismatch for coda /s/—that is, to aspirate or delete /s/ in ways associated with Puerto Rican Spanish. The regression analysis reveals a positive association between favorable national alignment and coda /s/ weakening ($\beta = 6.00$, $p = 0.01$), while the "indifferent" category also trended in the same direction, though less robustly ($\beta = 2.42$, $p = 0.10$). This suggests that speakers who express alignment—or at least lack resistance—to Puerto Rican identity are more likely to reproduce canonical dialectal features. A similar pattern emerged among Dominican speakers. While the "like" category was not significant on its own ($\beta = 0.71$, $p = 0.21$), indifference was strongly predictive of mismatch ($\beta = 1.22$, $p = 0.00$), suggesting that even a neutral stance toward national identification correlates with dialect maintenance. Taken together, these findings confirm that coda /s/ is a socially powerful resource. It is not simply preserved due to habit or exposure—it is mobilized in ways that correlate with national pride and in-group alignment. This reinforces earlier qualitative findings (see Chapter 4), where speakers like Pacífica, Danilo, and Pilar articulated clear associations between /s/ retention or loss and the distinctiveness of different varieties.

In contrast, tap mismatch—especially lateralization—functions as a more context-dependent and speaker-specific resource. For Puerto Ricans, liking PR identification was a strong predictor of tap lateralization ($\beta = 1.42$, $p < 0.01$), whereas indifference corresponded to lower rates of tap lateralization ($\beta = -0.33$, $p = 0.20$). This suggests that for some Puerto Rican speakers, lateralization is consciously embraced as a marker of identity, especially when positively aligned with their national group. Among Dominicans, both the "like" and "indifferent" categories were significantly associated with higher rates of mismatch—that is, producing the tap as [l]. Speakers who liked being identified as Dominican were strongly predictive of mismatch ($\beta = 4.80$, $p < 0.01$), as were those who expressed indifference ($\beta = 3.24$, $p < 0.01$). This pattern suggests that Dominican speakers actively mobilize tap lateralization as an identity marker, with strong effects for both positive and neutral stances toward national identification. This indicates that /ɾ/ lateralization is a socially recognized feature of Dominican Spanish—one that is acceptable, even expected, across a range of ideological positions. This interpretation is supported by qualitative data from speakers like Pilar and Danilo, who describe liquid variation as a hallmark of regional and national speech patterns.

These results reinforce the idea that salience plays a crucial role in shaping how linguistic features behave in contact settings—not by inherently preserving them, but by making them more available for social work. Highly salient features give speakers the option to make ideologically motivated choices: to maintain, reject, or strategically manipulate them depending on the context. This is evident in the treatment of rhotics, which many speakers preserve as dialectal markers despite long-term exposure to English and contact with other non-Caribbean varieties of Spanish. In contrast, features that are less perceptually salient (such as pronouns, subject position, and filled pauses) or carry weaker social meaning (such as lateral variation) are more prone to shift, often aligning more closely with English norms. These patterns support the view that salience is not an inherent property of linguistic forms but a socially mediated one, shaped by ideologies, experience, and identity work (Eckert 2008; Mendoza-Denton 2008). Ultimately, the data suggest that speakers make strategic linguistic decisions—sometimes preserving forms to assert in-group belonging, other times converging toward more neutral variants—navigating the dual pressures of bilingual optimization and social positioning.

7.2.2 Contextualizing the Findings: A Non-Generalizable Pattern

An important conclusion of this study is that these patterns are not universally applicable. Different contact groups, shaped by distinct linguistic loyalties, sociolinguistic ideologies, and historical experiences, will exhibit varying degrees of salience and may treat even highly salient features differently. For instance, communities prioritizing integration may demonstrate leveling in high-salience features, while those emphasizing differentiation, may preserve dialectal distinctiveness. This variability highlights the importance of contextualizing salience within the sociolinguistic and cultural dynamics of individual communities.

The findings align closely with Erker & Vidal-Covas's (2024) analysis of Spanish in Boston, which demonstrates that contact-induced change is not simply about structural convergence or attrition, but about the social meaning of linguistic features. In that study, the authors observed that socially weak variables undergo systematic convergence with English norms across generations. In contrast, socially strong variables remain stable or even become more entrenched over time. Crucially, this pattern was not framed as a linguistic inevitability but as an outcome of the sociolinguistic orientation of the community.

Indeed, the Boston case represents a context in which speakers show strong dialectal loyalty and a preference for maintaining nationally recognizable speech patterns. As Erker & Vidal-Covas argue, contact does not simply flatten difference—it can also amplify the salience of already prominent features. In multilingual urban centers like Boston, where speakers of multiple dialects coexist, socially strong features may become even more powerful stylistic resources, enabling speakers to assert in-group belonging and resist dialectal leveling.

However, this is not a universal pattern. In other contexts—such as Salvadoran and Honduran communities in the Pacific Northwest—speakers have been shown to reduce the use of highly salient features (such as voseo) as a strategy of integration into predominantly Mexican-American speech communities (Woods & Rivera-Mills 2012; Sorenson 2013). In such cases, the same types of features that remain stable in Boston (e.g., /s/ weakening or liquid variation) are sites of change elsewhere. The difference lies not in the linguistic features themselves, but in the social value attributed to them by the speakers.

This underscores a critical point: salience is not an inherent or static property of a linguistic form. Instead, it is a contextually situated phenomenon—shaped by language ideologies, patterns of intergroup contact, and the social goals of speakers. As such, the outcomes of contact cannot be generalized across communities. Rather, they must be interpreted through the lens of local sociolinguistic dynamics, including how speakers perceive and mobilize particular forms as resources for identity, inclusion, or differentiation.

In this light, the findings presented here should be seen not as a blueprint for contact-induced change, but as one instantiation of how sociolinguistic salience, speaker agency, and language contact interact in a specific community. Future work would benefit from further comparative analyses across contact settings, with close attention to the social meanings that linguistic forms carry—and how those meanings are shaped by local histories, ideologies, and interactional practices.

7.3 Conclusion and Future Directions

This dissertation has argued that variation in bilingual contact settings cannot be fully understood without attending to how speakers use linguistic features to navigate social meaning. Salient forms do not behave differently because they are intrinsically more stable or resistant to change. Rather, they are treated differently by speakers—mobilized strategically, preserved ideologically, or reconfigured socially—depending on local histories, community orientations, and the speaker's position within them.

This study highlights the complex interplay between linguistic salience, contact-induced change, and sociolinguistic agency. Among the variables analyzed here, lower-salience features pattern in ways that reflect bilingual optimization strategies, converging toward English norms in predictable and systematic ways. It is worth noting, however, that while the statistical associations between contact experience and the use of lower-salience features are significant, their effect sizes are modest. These patterns point to a meaningful trend toward convergence, but they should not be interpreted as wholesale structural shift. A more fine-grained, token-level analysis of the linguistic factors conditioning these features would be necessary to fully assess the extent and nature of convergence. This underscores the importance of interpreting statistical results in light of broader linguistic patterns and interactional practices.

By contrast, highly salient features are subject to social pressures that vary across speakers and settings. They are sometimes maintained, sometimes reconfigured, and sometimes attenuated, depending on how they are ideologically positioned by speakers. In this sense, the key insight is not that salience leads to stability, but that salience decouples linguistic features from the cognitive pressures that drive convergence. As a result, salient features are more available for stylistic manipulation—but also more contingent, more context-sensitive, and less predictable.

The patterns observed in Boston, while informative, are shaped by the specific sociolinguistic orientations of this community—particularly the value placed on dialectal distinctiveness and linguistic maintenance. As discussed earlier, other communities may exhibit different orientations, leading to markedly different outcomes. This reinforces the broader point that salience alone does not determine contact-induced change. Rather, it is speakers' social goals, ideological positions, and community dynamics that guide how linguistic features are interpreted, evaluated, and deployed.

This study contributes to sociolinguistic theory by centering speaker agency in models of contact-induced change, emphasizing that variation is shaped not only by structural pressures but by socially meaningful choices. It demonstrates that salience is not a fixed or purely perceptual property, but one that emerges through folk commentary, identity work, and patterns of interaction. In doing so, it challenges models that treat convergence as an inevitable outcome of contact, showing instead that contact can also amplify socially meaningful forms—especially when speakers orient toward them as resources for differentiation or belonging.

These findings also open several paths for future research. Comparative work across communities could help reveal how the same linguistic features are treated differently by language users depending on local ideologies, histories, and patterns of interaction.

Perception-based studies, including matched-guise experiments or salience ranking tasks, could shed light on how speakers evaluate forms and whether their perceptions align with actual usage. Finally, social network analysis and ethnographic methods offer promising tools for understanding how speakers' affiliations, relationships, and lived experiences shape their linguistic behavior over time.

In addition to these broader directions, future research could also expand the methodological approach to phonological analysis. While the match/mismatch framework[1] used here may simplify some of the finer-grained distinctions between variants, it enables a more inclusive analysis by capturing any surface realization that diverges from the phonological form predicted by distributional context. Some prior studies have grouped variants into broader categories—such as "weakening" or "lateralization"—allowing for focused analyses across specific subsets of variation. However, the approach adopted here offers a unified framework that accommodates a wider range of phonetic outcomes, including less frequent or less-categorically analyzed realizations, providing a more comprehensive view of the variation that occurs in naturalistic speech. Together, these approaches can deepen our understanding of how language variation operates not just as a system of rules, but as a terrain of social meaning navigated by real people in real communities.

7.3.1 Final Thoughts

Linguistic features do not behave—*speakers* do. This dissertation has shown that when it comes to outcomes of contact, change and/or stability are not random, nor are they in-

[1]The use of terms like "predicted" or "mismatch" in this study is not intended to imply that certain variants are incorrect. Rather, these terms are used descriptively to reflect surface realizations that differ from the phonological form associated with a given lexical or distributional context. This framework is analytic, not prescriptive, and is motivated by a commitment to documenting and valuing the full range of linguistic variation as it naturally occurs.

evitable. They reflect speaker decisions—conscious or not—about how to sound, who to align with, and what social meanings to project. Salience, then, is not just about perception or frequency. It is about value. And that value is shaped by context: by the histories speakers inherit, the ideologies they reproduce or resist, and the social terrain they move through. Understanding contact-induced change means understanding how speakers *navigate that terrain*—sometimes adapting, sometimes differentiating, always making choices. The case of Spanish in Boston provides one map. It shows how bilingual speakers manage competing pressures: structural convergence, identity expression, inter-dialectal contact, and community expectations. But it is only one map. Other communities, other contexts, will draw their linguistic landscapes differently. And that, too, is the point.

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